POLITECNICO
MILANO 1863

Adaptative and Nonlinear Control
UAV Project

1. Model introduction

POLITECNICO
MILANO 1863

- **UAV Dynamics**

$$M\dot{\nu}_b + \nu_b^{\times} M\nu_b = -D\nu_b + w_g + Fu$$

- **Propeller Dynamics**

$$\dot{\Omega}_r = -k_{r,m}\left(\Omega_r - \Omega_{r,cmd}\right) \qquad r = 1, \dots, 6$$

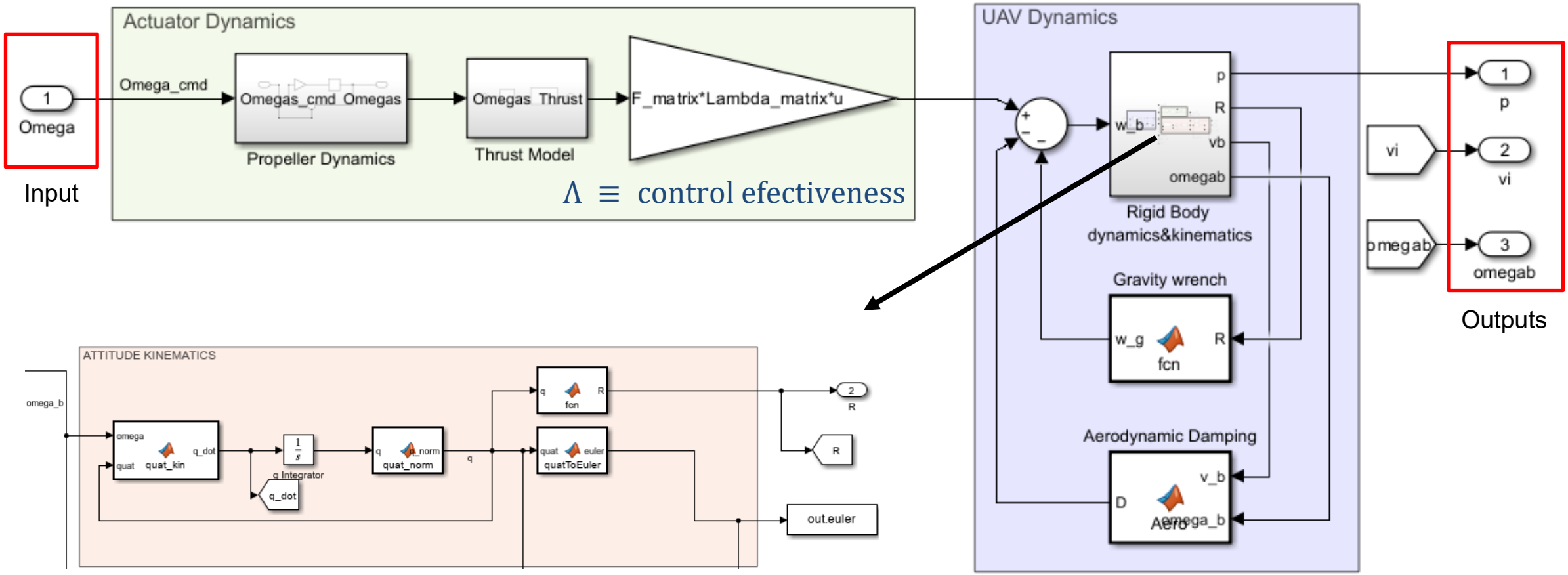
$$u^T = [T_1 \quad \dots \quad T_{\mathrm{N}}]$$

$$T_r = \mathrm{sat}_{T_m}^{T_M}\left(\lambda_r k_f \Omega_r^2\right)$$

- **Allocation**

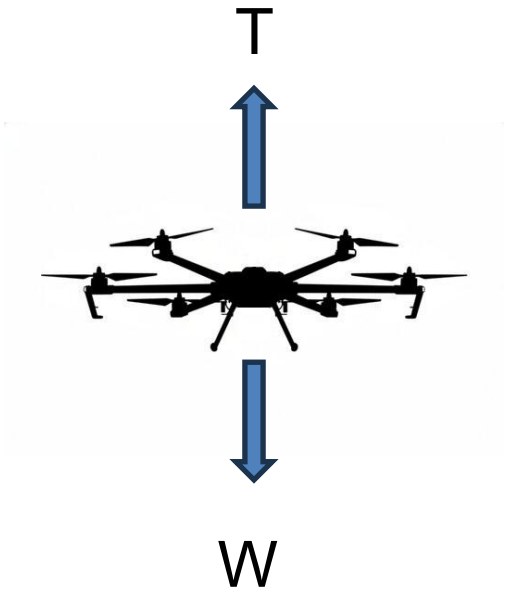
$$\begin{pmatrix} \mathrm{T_c} \\ \tau_{\mathrm{c}} \end{pmatrix} = \mathrm{Fu}$$

1.1. Simulink Model



- Use of quaternions and rotation matrices to avoid Euler angles singularities

1.2. Model Trim
Static Position Trim



Trim

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{u})$$

$$\dot{\boldsymbol{x}} = \boldsymbol{0} \rightarrow f(\boldsymbol{x}_{trim}, \boldsymbol{u}_{trim}) = 0$$

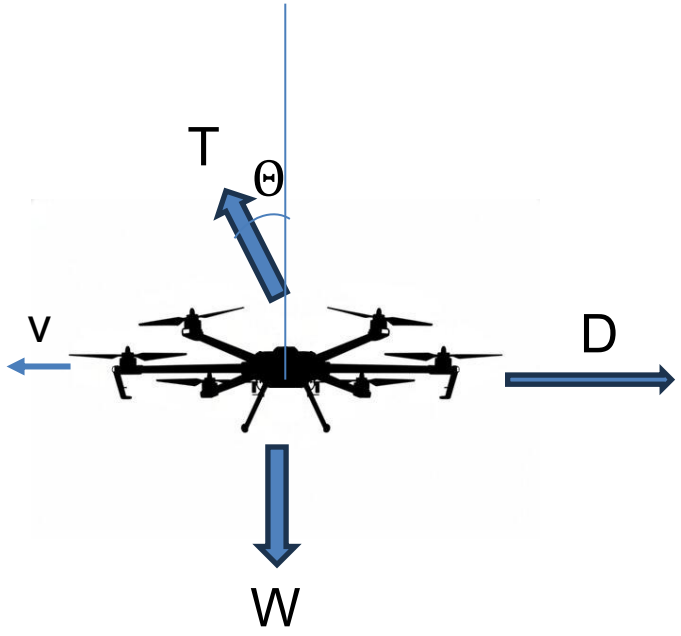
$$\boldsymbol{A} = \left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{\boldsymbol{x}_{trim}, \boldsymbol{u}_{trim}} \quad ; \quad \boldsymbol{B} = \left.\frac{\partial f}{\partial \boldsymbol{u}}\right|_{\boldsymbol{x}_{trim}, \boldsymbol{u}_{trim}} \rightarrow \dot{\boldsymbol{x}} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{B}\boldsymbol{u}$$

Simple trim case

$$\boldsymbol{p} = [0,0,1]^T \quad // \quad \boldsymbol{v} = \boldsymbol{\omega}_{\boldsymbol{b}} = \boldsymbol{0}$$

$$T = 6k_f\Omega^2 = mg \quad \rightarrow \quad \Omega = \sqrt{\frac{mg}{6k_f}} = 733.16 \text{ rad/s}$$

1.2. Model Trim
Constant x velocity



$$\boldsymbol{v} = [1,0,0]^T // \boldsymbol{\omega_b} = \mathbf{0}$$

- In this case, $\boldsymbol{p}$ is not constant. Since any internal force of the system is $\mathrm{f(p)}$, it can not be taken into account when solving:

$$\dot{\boldsymbol{x}} = \frac{d}{dt}\begin{pmatrix} \boldsymbol{v} \\ \boldsymbol{\omega_b} \end{pmatrix} = \mathrm{f}(\boldsymbol{x}, \boldsymbol{u}) = \mathbf{0}$$

- Now the equation is solved with minimizing f with constrains in the previously known values of $\boldsymbol{v}$ & $\boldsymbol{\omega}$.

	Rad/s
Ω_1	730.5
Ω_2	734.8
Ω_3	739.7
Ω_4	730.6
Ω_5	728.8
Ω_6	734.0

2. Baseline controller implementation, tuning and testing

2.1 Control architecture for position-heading tracking control

Position Controller

INPUT	OUTPUT
1. Desired trajectory $\boldsymbol{p_d}, \boldsymbol{v_d}, \boldsymbol{a_d}$ 1. Current state $\boldsymbol{p}, \boldsymbol{v}$	1. Control force required $\bar{\boldsymbol{f}}_c$

Allocation

INPUT	OUTPUT
1. Desired forces and moments in body reference $\bar{\boldsymbol{f}}_c, \boldsymbol{\tau_c}$	1. Ω commanded for each propeller

Attitude Planner

INPUT	OUTPUT
1. Control force required $\bar{\boldsymbol{f}}_c$ 2. Heading ψ	1. Desired rotation matrix $\boldsymbol{R_d}$

Attitude Controller

INPUT	OUTPUT
1. Actual and desired rot. matrix $\boldsymbol{R_d}, \boldsymbol{R}$ 2. Body rotational velocity $\boldsymbol{\omega_b}$	1. Desired control moments $\boldsymbol{\tau_c}$

2.1 Control architecture for position-heading tracking control
Time scales separation

- Properly assigning **characteristic times** for each step of the control architecture is needed to ensure a correct behavior of the system.
- In the **linear range** of a first order system, it converges in approximately $\mathbf{3\tau}$, with τ being the characteristic time. With this number in mind, it can be successively estimated the characteristic time for each control step.

Actuator Dynamics

$\tau_{act} = 0.05s$

Attitude Control

$\tau_{att} \approx 3 - 5\ \tau_{act}$

$\tau_{att} \approx 0.2\ \mathrm{s}$

Velocity Control

$\tau_{\mathrm{vel}} \approx 3 - 5\ \tau_{att}$

$\tau_{\mathrm{vel}} \approx 0.6\ \mathrm{s}$

Position Control

$\tau_{\mathrm{pos}} \approx 3 - 5\ \tau_{\mathrm{vel}}$

$\tau_{\mathrm{pos}} \approx 2\ \mathrm{s}$

2.1. Control architecture for position-heading tracking control
Velocity and position control architecture chosen

Standard PID or cascade?

Suppose a simplified equation without disturbances:

$$\dot{v} = f_c$$

$$\begin{cases} \dot{\mathrm{v}} = f_c = -K_p e_p - K_v e_v - K_i e_i \\ e_v = \dot{e}_p; \dot{e}_i = e_p \end{cases}$$

Lyapunov function proposal

$$V = 1/2 K_p e_p^2 + 1/2 e_{\mathrm{v}}^2 + 1/2 K_i e_i^2$$

$$\dot{V} = K_p e_p e_v + e_v \dot{e}_v + K_i e_i e_p = -K_{\mathrm{v}} e_v^2 + K_i e_i (e_p - e_v)$$

- Cross terms $K_i e_i (e_p - e_v)$ leads to Lyapunov function being not **negative semi-definite**. To prove stability, **Lasalle principle** or **cross terms Lyapunov function** is needed.
- Position and velocity gains are **coupled**, leading to practical tuning problems and bandwidth overlapping.

2.1. Control architecture for position-heading tracking control
Velocity and position control architecture chosen

Standard PID or <u>cascade</u>?

$$\begin{cases} f_c = -K_{\text{vel}}(v - v_{cmd}) - K_i\xi & \Longrightarrow \text{Internal} \\ v_{cmd} = K_{\text{pos}}(p_{cmd} - p) & \Longrightarrow \text{External} \\ \dot{\xi} = (v - v_{cmd}) = \hat{v} \end{cases}$$

- **Both** internal and external loops **asymptotically stable**. Easier tunning, no cross dependence.

Suppose a simplified equation without disturbances:

$$\dot{v} = f_c$$

$$V_{int} = 1/2\hat{v}^2 + 1/2K_i\xi^2$$

$$\dot{V}_{int} = -K_{vel}\hat{v}^2 < 0$$

Asymptotically Stable

For external control, consider internal instantaneous ($\tau_{pos} \approx 3 - 5\tau_{vel}$):

$$V_{\text{ext}} = 1/2e_p^2 \rightarrow \dot{\text{V}}_{ext} = e_p\dot{e}_p = -K_{\text{pos}}e_p^2 < 0$$

Asymptotically Stable

2.1. Control architecture for position-heading tracking control
Attitude Planner and Control

$$\mathbf{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_\mathbf{c}$$

Planner

- Define $\boldsymbol{k_d}, \boldsymbol{j_d}$& $\boldsymbol{i_d}$ and therefore R_d.

1. $\mathbf{k_d} = \frac{\mathbf{f_c}}{|\mathbf{f_c}|}$
2. $\mathbf{j_d} = \frac{\mathbf{k_d} \times \mathbf{h_d}}{|\mathbf{k_d} \times \mathbf{h_d}|}, \mathbf{h_d} = [\cos\psi_\mathrm{d}, \sin\psi_\mathrm{d}, 0]^\mathrm{T}$
3. $\mathbf{i_d} = \mathbf{j_d} \times \mathbf{k_d}$

$$\Downarrow$$

$$R_d = (\boldsymbol{i_d}, \boldsymbol{j_d}, \boldsymbol{k_d})$$

Control

- $R = (\boldsymbol{i_b}, \boldsymbol{j_b}, \boldsymbol{k_b})$
- Define orientation error:

$$\mathbf{e_R} = \frac{1}{2}\left(\boldsymbol{R_d^T R} - \boldsymbol{R^T R_d}\right)^\vee$$

- Lyapunov function:

$$\mathcal{V} = \frac{1}{2} tr\left(\mathrm{I} - \boldsymbol{R_d^T R}\right) + \frac{1}{2}\boldsymbol{\omega^T J \omega}$$

- Proposed controller:

$$\boldsymbol{\tau}_c = -K_R \mathbf{e_R} - K_\omega \boldsymbol{\omega}$$

- Substituting:

$$\dot{\mathcal{V}} = -\boldsymbol{\omega^T} K_\omega \boldsymbol{\omega} \leq 0$$

Lyapunov stable, Almost Globally Asymptotically Stable

2.1. Control architecture for position-heading tracking control
Allocation

1. $\begin{pmatrix} \boldsymbol{T_c} \\ \boldsymbol{\tau_c} \end{pmatrix} = \boldsymbol{F u_c}$

2. $\boldsymbol{u_c} = \begin{bmatrix} T_{1,cmd} & \dots & T_{N_r,cmd} \end{bmatrix}^{\top}$

3. $T_{\mathrm{i,cmd}} = k_f \Omega_{\mathrm{i},cmd}^2$

4. $\Omega_{i,cmd} \xrightarrow{\text{Actuator Dynamics}} \Omega_i$

5. $saturation(\Omega_{\mathrm{i}}) \longrightarrow T_{1,\mathrm{act}}$

6. $\begin{pmatrix} \boldsymbol{T} \\ \boldsymbol{\tau} \end{pmatrix} = \boldsymbol{F u}_{\mathrm{act}}$

F SVD decomposition

$$\boldsymbol{M} = \boldsymbol{U \Sigma V}^T$$

$\boldsymbol{V}^T =$

-0.408	-0.408	-0.408	-0.408	-0.408	-0.408	Vertical
0.000	-0.500	-0.500	0.000	0.500	0.500	Roll
-0.577	-0.289	0.289	0.577	0.289	-0.289	Pitch
-0.408	0.408	-0.408	0.408	-0.408	0.408	Yaw
-0.533	0.459	0.074	-0.533	0.459	0.074	Null space
0.222	0.350	-0.573	0.222	0.350	-0.573	

$\boldsymbol{\Sigma} =$

2.449	0.000	0.000	0.000	0.000	0.000
0.000	0.372	0.000	0.000	0.000	0.000
0.000	0.000	0.372	0.000	0.000	0.000
0.000	0.000	0.000	0.220	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000
0.000	0.000	0.000	0.000	0.000	0.000

$\boldsymbol{U} =$

0.000	0.000	0.000	0.000	-0.899	-0.437
0.000	0.000	0.000	0.000	-0.437	0.899
-1.000	0.000	0.000	0.000	0.000	0.000
0.000	-1.000	0.000	0.000	0.000	0.000
0.000	0.000	1.000	0.000	0.000	0.000
0.000	0.000	0.000	1.000	0.000	0.000

2.2. Control architecture for position-heading tracking control
Attitude Controller Tuning – Roll & Pitch

Objectives

- $3(3\tau_{act}) \approx \mathbf{0.45\ s}$, approximate **settling time**.
- **Small overshoot**, attitude needs to be smooth.
- Nearly 2 time faster at Roll & Pitch than Yaw.
- Propeller's **saturation** considerations.

$$\tau_x \,\&\, \tau_y < 1.6\ \mathrm{Nm}$$

Sensors & Actuators

- $f_c \approx 20\ \mathrm{rad/s}$
- $\tau_{delay} = 0.002\ \mathrm{s}$

Simulated as first order system

Simplified Attitude Model

$$\begin{pmatrix} \dot{R} = R\omega_b^{\times} \\ J\dot{\omega}_b = -\omega_b^{\times} J\omega_b + \tau_c - D\omega_b \end{pmatrix}$$

For simplicity, small angular velocity and small angles are considered, leading to:

$$\ddot{\Theta} + \frac{D_x + K_{vx}}{J_{xx}}\dot{\Theta} + \frac{K_{rx}}{J_{xx}}\Theta = 0$$

$$\tau_{att} = \frac{1}{\xi\omega_n} \approx 0.20 \longrightarrow \begin{cases} K_{rx} \approx J_{xx}\omega_n^2 \\ K_{vx} \approx 2J_{xx}\xi\omega_n \\ \xi_{des} = 0.8 \end{cases} \quad (\omega_n \approx 6.3\ \mathrm{rad/s})$$

Overshoot of 0.15%
Settling time of 0.45 s

2.2. Control architecture for position-heading tracking control
Attitude Controller Tuning - Yaw

Objectives

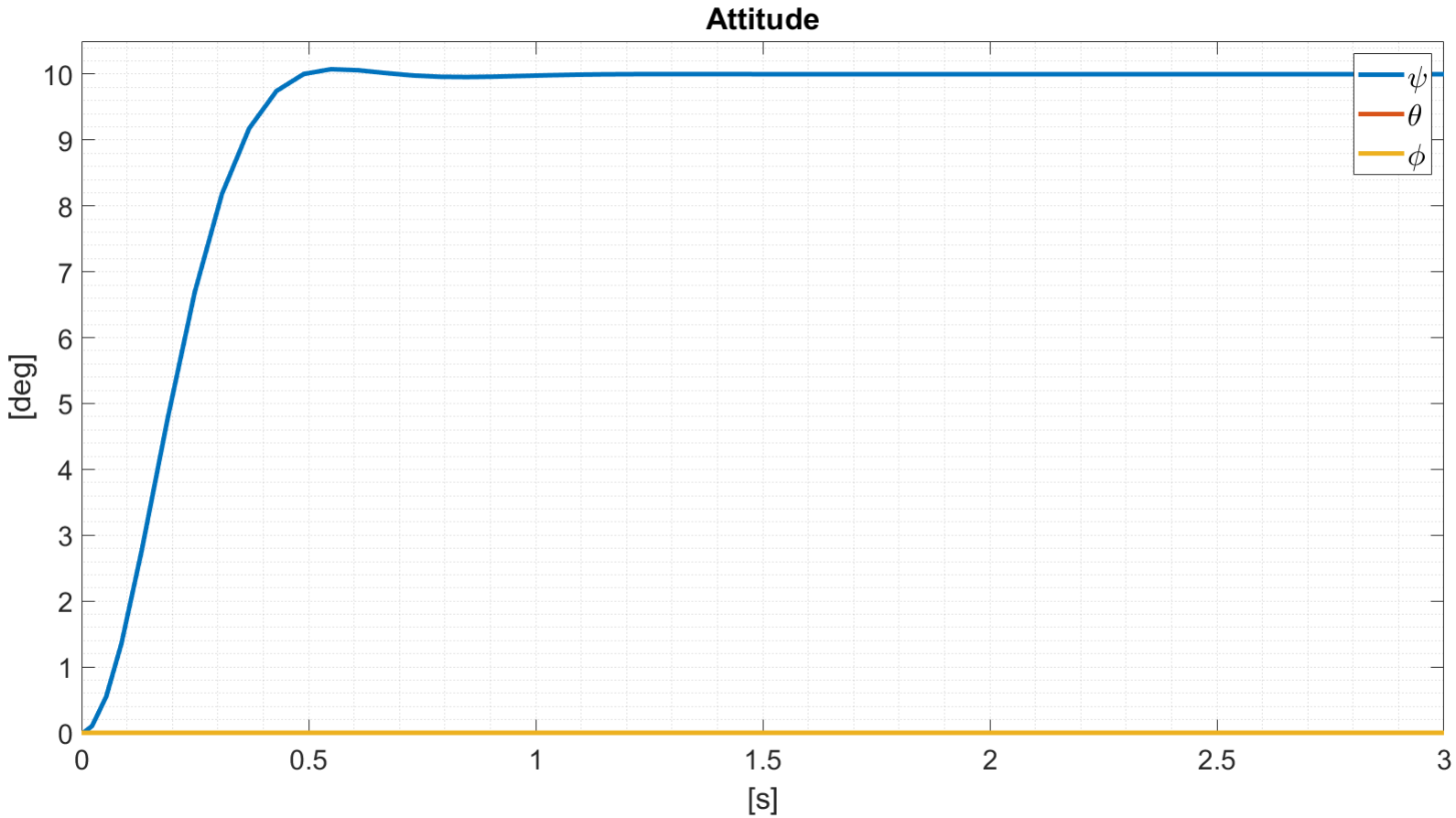
- Small overshoot. Yaw half as fast as roll and pitch, to **prevent saturation**.
- $2\tau_{\mathrm{roll}} \approx 1\ \mathrm{s}$, approximate **settling time**
- Propeller's saturation considerations.

$$\tau_{\mathrm{z}} < 1.13$$

- Same process as before, it leads to:

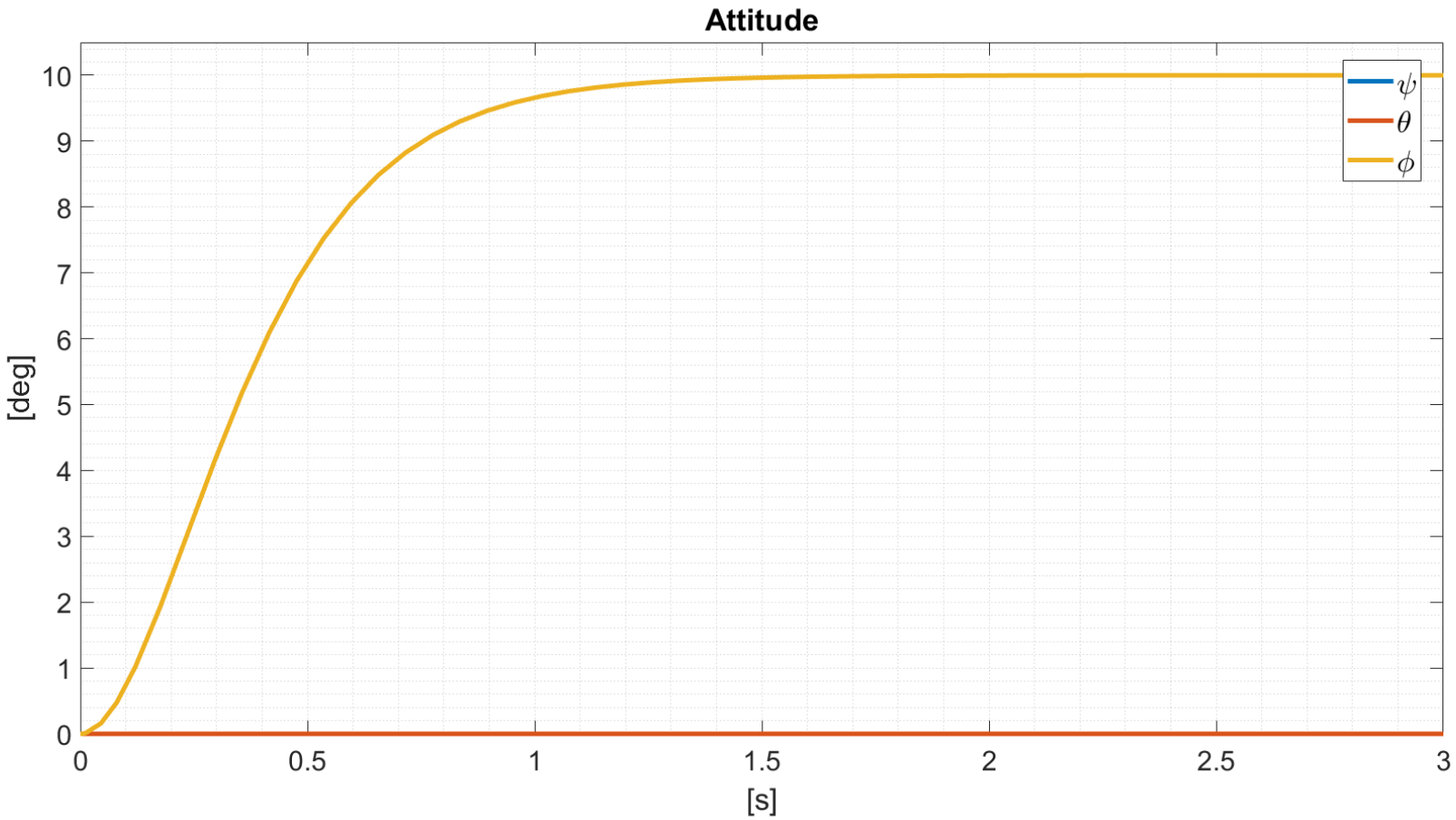
$$\xi \approx 0.9\,, \omega_n \approx 4$$

2.2. Control architecture for position-heading tracking control
Roll & Pitch attitude control behavior – simplified model



- Roll expected behavior $(\xi = 0.8, \omega_n = 6.25 rad/s)$
- Settling time (2%) ≈ 0.44
- Overshoot $= 0.08\ \%$

2.2. Control architecture for position-heading tracking control
Yaw attitude control behavior – simplified model



- Yaw behavior ($\xi = 0.9, \omega_n = 4\, rad/s$)
- Settling time (2%) $\approx 1.1\ s$
- Overshoot $\approx\ 0\ \%$

2.2. Control architecture for position-heading tracking control
Velocity Control – X & Y

Objectives

- $\tau_{vel} \approx 0.45$
- $(3\tau_{vel}) \approx 1.5$ s, approximate **settling time** of XY velocity. Needed for position control to be this fast.

Sensors & Actuators/Attitude

- $\tau_{delay,\text{sensor}} = 0.02$ s
- $\tau_{delay,att} = 0.15$ s

  Simulated as first order system

Simplified Attitude Model

$$\dot{\boldsymbol{v}} = -D\boldsymbol{v}/\mathrm{m} - \boldsymbol{g} + \boldsymbol{f}_c/m$$

$$\boldsymbol{f}_c = -K_v(\boldsymbol{v} - \boldsymbol{v}_{cmd}) - K_i\boldsymbol{e}_i$$

$$\dot{\boldsymbol{e}}_i = (\boldsymbol{v} - \boldsymbol{v}_{cmd})$$

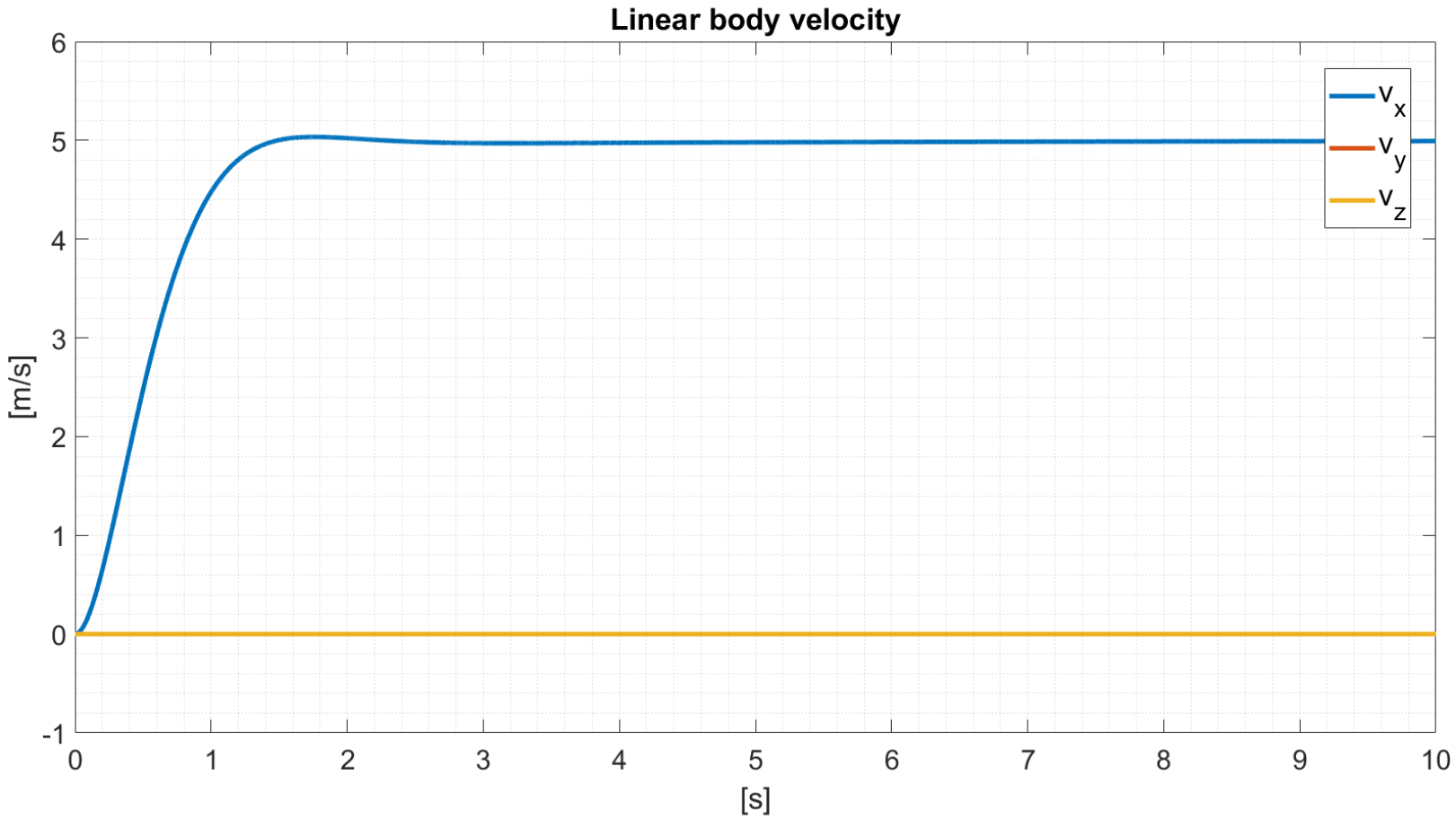
$$\ddot{\boldsymbol{v}} + \frac{D + K_v}{\mathrm{m}}\dot{\boldsymbol{v}} + \frac{K_i}{m}\boldsymbol{v} = \frac{K_i}{m}\boldsymbol{v}_{cmd}$$

$$\omega_n \approx 1 \text{ rad/s}$$

$$\xi \approx 1$$

- $K_i \approx m\omega_n^2$
- $K_\mathrm{v} \approx m2\xi\omega_n$

2.1. Control architecture for position-heading tracking control
Velocity Control step response - XY



- X velocity step behavior $K_v = 0.7$ & $K_i = 3.8$
- Settling time (2%) $\approx 1.4\ s$
- Overshoot $\approx\ 1\ \%$

2.2. Control architecture for position-heading tracking control
Velocity Control – Z

Objectives

- $3(3\tau_{att}) \approx 1.5\ \mathrm{s}$, approximate **settling time** of Z position.
- However, in order to quickly adapt the Z position in a forward flight inclination, and due to the **natural orientation of the propeller's force**, it should be a bit **faster**.

$$0.38\ \mathrm{N} < T_Z < 25.5\ \mathrm{N}$$

Sensors & Actuators/Attitude

- $\tau_{delay,\mathrm{sensor}} = 0.02\ \mathrm{s}$
- $\tau_{delay,act} = 0.05\ \mathrm{s}$

Simplified Attitude Model

$$\dot{\boldsymbol{v}} = -\boldsymbol{g} + \boldsymbol{f}_c/m$$

$$\boldsymbol{f}_c = -K_v(\boldsymbol{v} - \boldsymbol{v}_{cmd}) - K_i \boldsymbol{e}_i$$

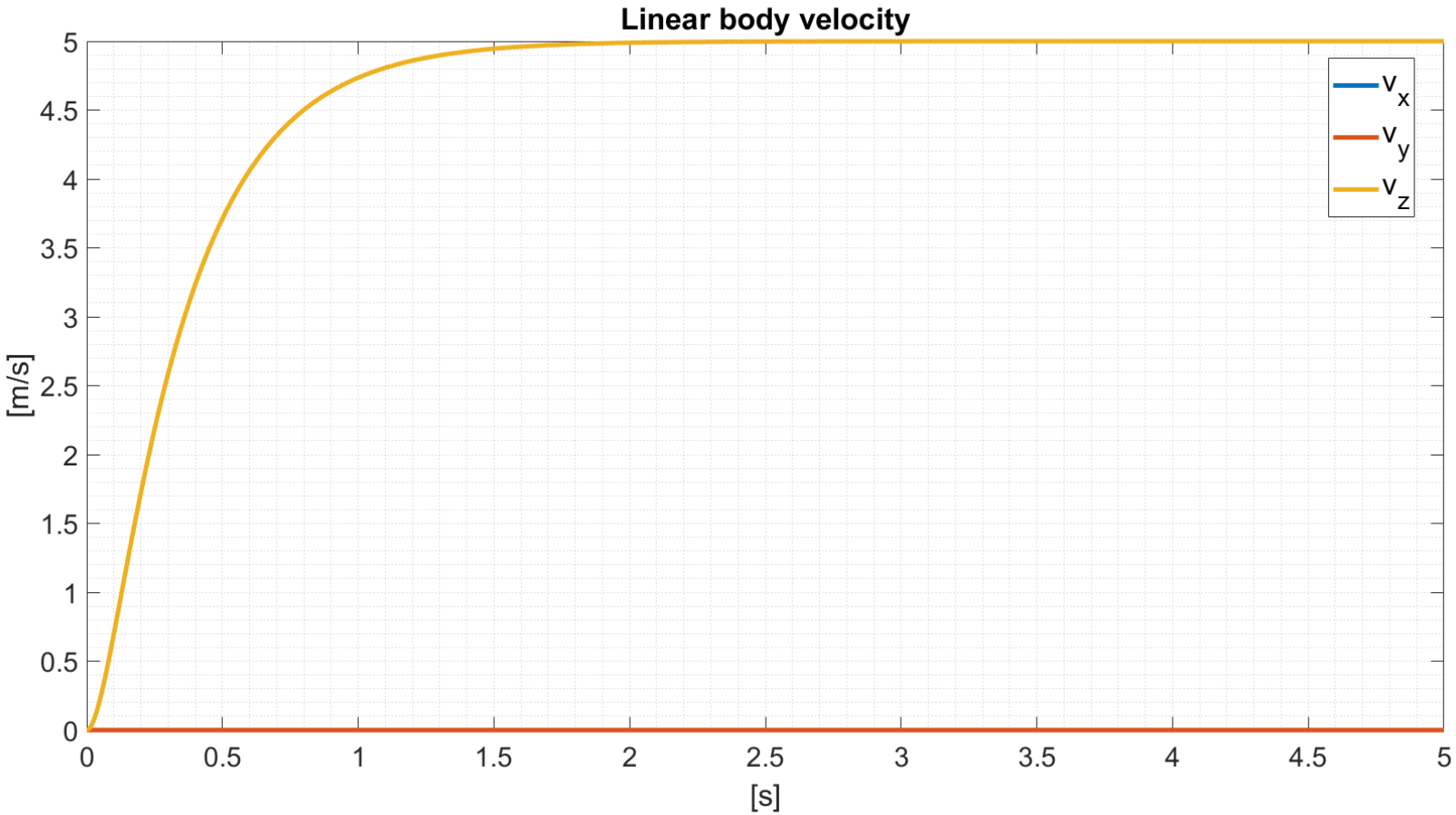
$$\dot{\boldsymbol{e}}_i = (\boldsymbol{v} - \boldsymbol{v}_{cmd})$$

$$\ddot{\boldsymbol{v}} + \frac{K_v}{\mathrm{m}}\dot{\boldsymbol{v}} + \frac{K_i}{m}\boldsymbol{v} = \frac{K_i}{m}\boldsymbol{v}_{cmd}$$

If Ki = 0, it results in an integrator (first order system). Without uncertainties in mass or vertical bias perturbations, and with weight feedforward, no need for it.

$$K_v = 3; K_i = 0$$

2.1. Control architecture for position-heading tracking control
Velocity Control step response - z



- Z velocity step behavior $K_v = 3$ & $K_i = 0$
- Settling time (2%) ≈ 1.3
- Overshoot $=$ 0 %

2.2. Control architecture for position-heading tracking control
Position Control

Objectives

- $\tau_{pos} \approx 3\tau_{\text{vel}} \approx 1.5s$
- **Settling time** of around 4.5 s

$\tau_{pos,\text{settling}} = 4/K_{pos,\text{xy}}$; $\tau_{pos,\text{settling}} \approx 4\tau_{pos} \approx 4.5\ s$

$K_{pos,\text{xy}} \approx 0.9$

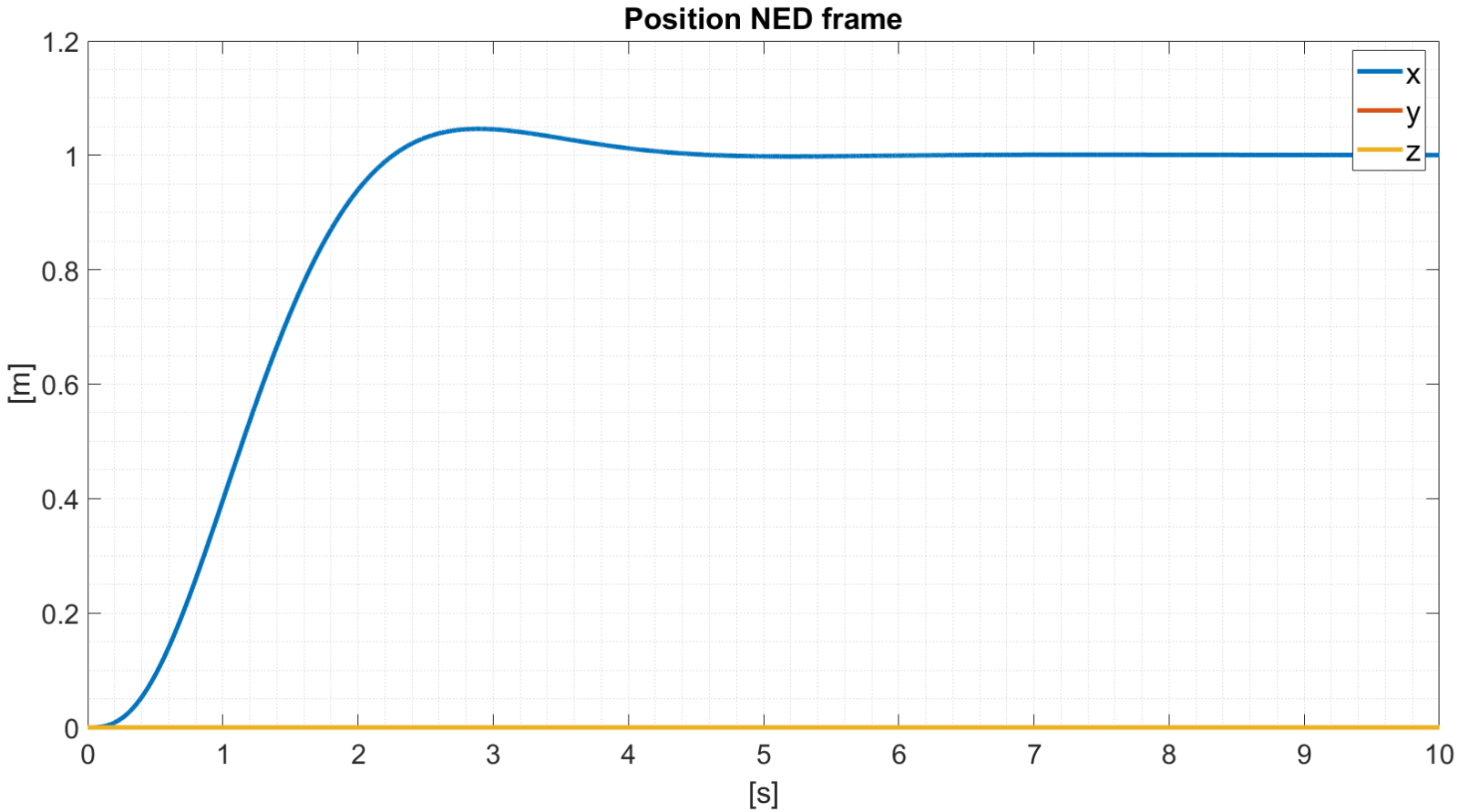
1. $P(s) = V(s)/s$
2. $E = P_{ref} - P$
3. $V_{cmd} = K_{pos}E$
4. $V_{cmd} \approx V$
5. $P(s) = K_{pos}(P_{cmd} - P)/s$
6. $\frac{P}{P_{cmd}} = \frac{K_{pos}}{s+K_{pos}}$

For Z, same with slightly smaller settling time, due to propellers orientation and gravity help.

$K_{pos,z} \approx 1$

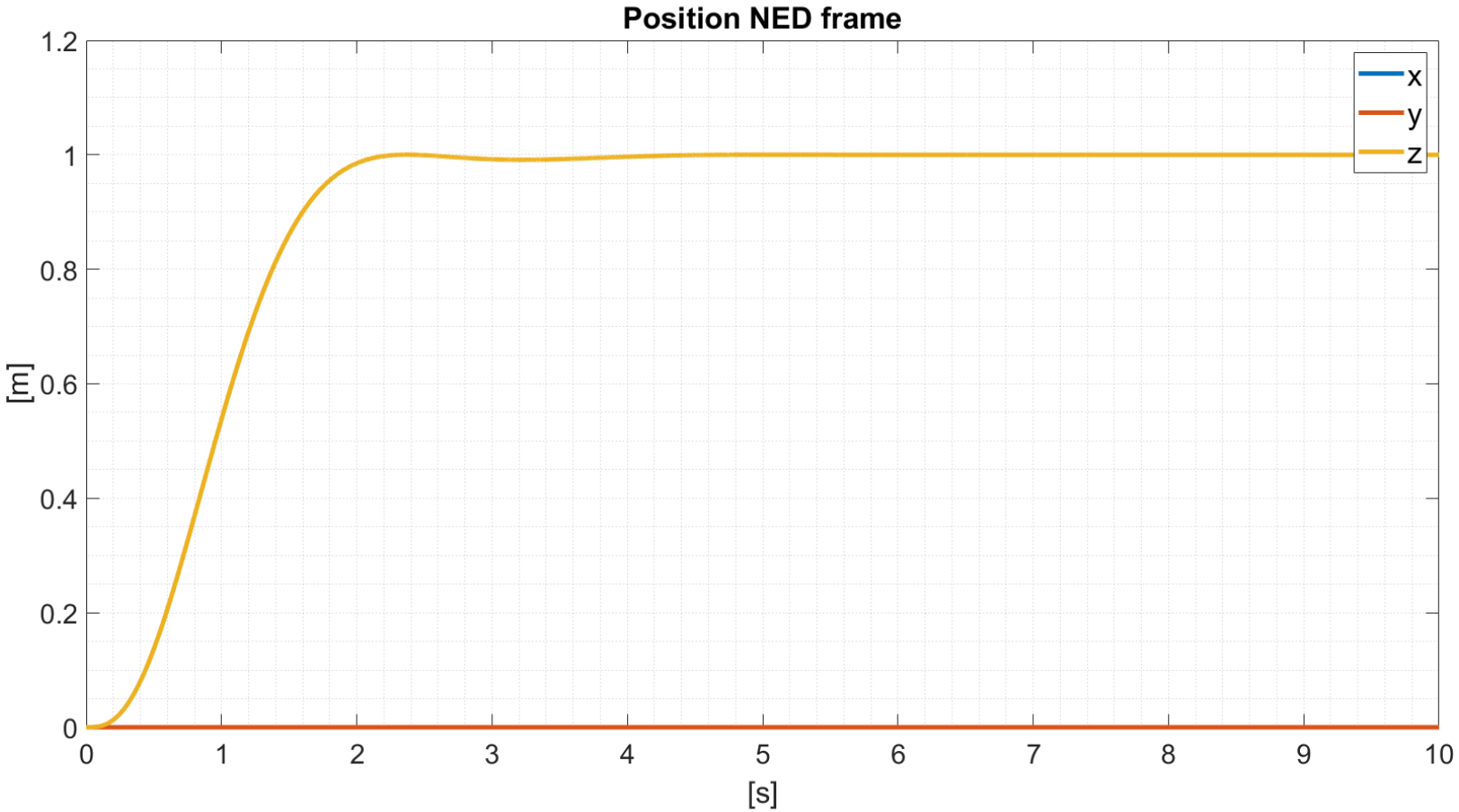
- First order system if velocity control considered instantaneous, no need for integral action

2.1. Control architecture for position-heading tracking control
Position Control step response - XY



- Z velocity step behavior $K_{pos} = 0.9$
- Settling time (2%) $\approx 3.8\ s$
- Overshoot $=\ 5\ \%$

2.1. Control architecture for position-heading tracking control
Position Control step response - Z



- Z velocity step behavior $K_{pos} = 1$
- Settling time (2%) $\approx 2\ s$
- Overshoot $\approx\ 0\ \%$

2.3. Control architecture for position-heading tracking control
Full model test

Tests

1. Simple XY & Z position step
2. 8 shape
3. Simple trajectory

- V max = 5 m/s

Sensors

- $\tau_{delay,att} = 0.002s$
- $\tau_{delay,pos} = 0.02$
- $Bandwitdh\ attitude\ \text{sensors} = 500\ rad/s$
- $Bandwitdh\ \text{position sensors} = 20\ rad/s$

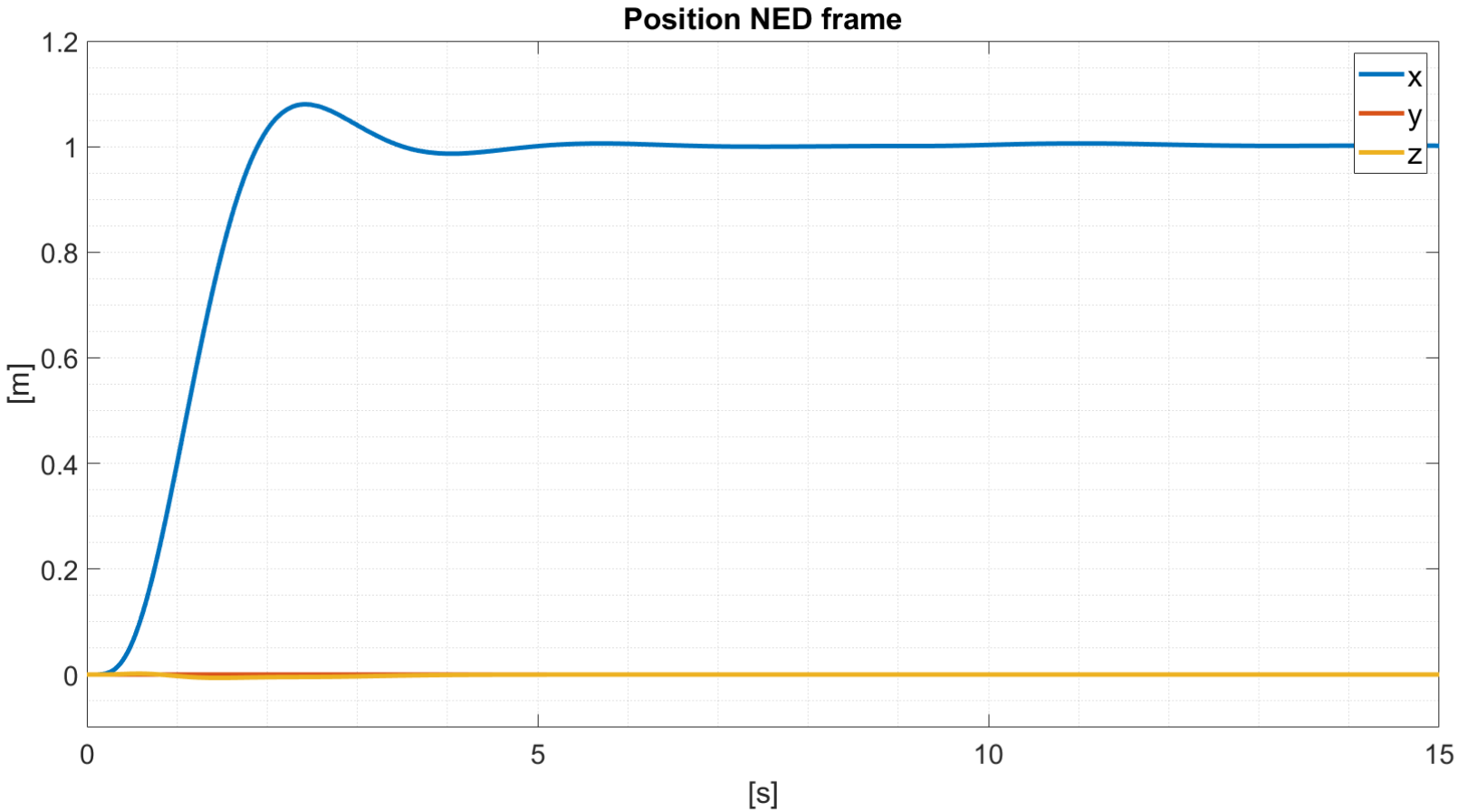
Metric used

- Root Mean Square Error

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(e_i)^2}$$

- Problem is by design the controller needs the reference to be $\approx V_{mean}/K_p$ meters ahead for the correct following

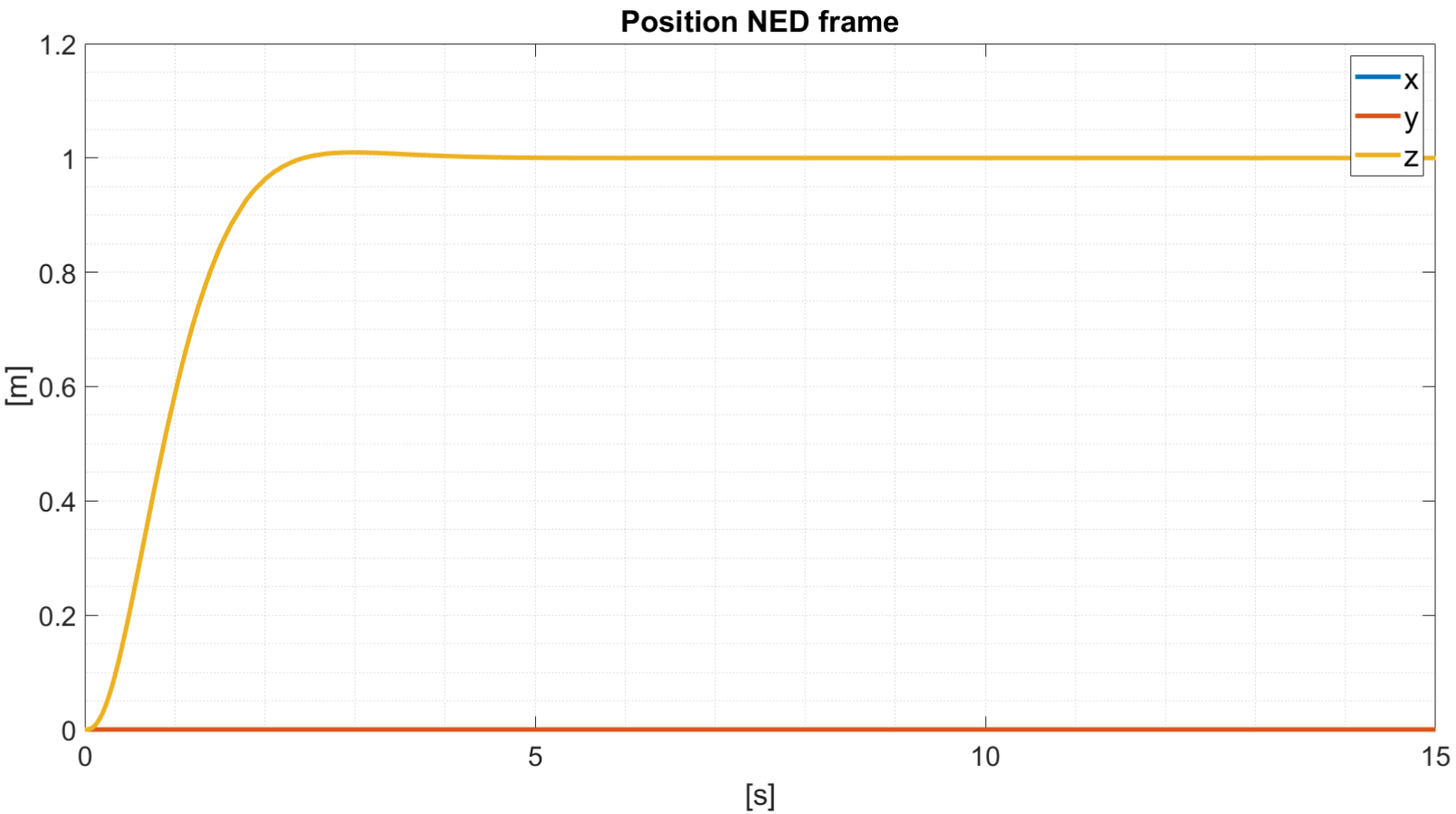
2.1. Control architecture for position-heading tracking control
Step response full model - XY



- Z velocity step behavior $K_{pos} = 0.9$
- Settling time (2%) $\approx 3.3\ s$ (Nominal 3.8 s)
- Overshoot $= 8\ \%$ (Nominal 5 %)

*Z fall of <1cm

2.1. Control architecture for position-heading tracking control
Step response full model - Z



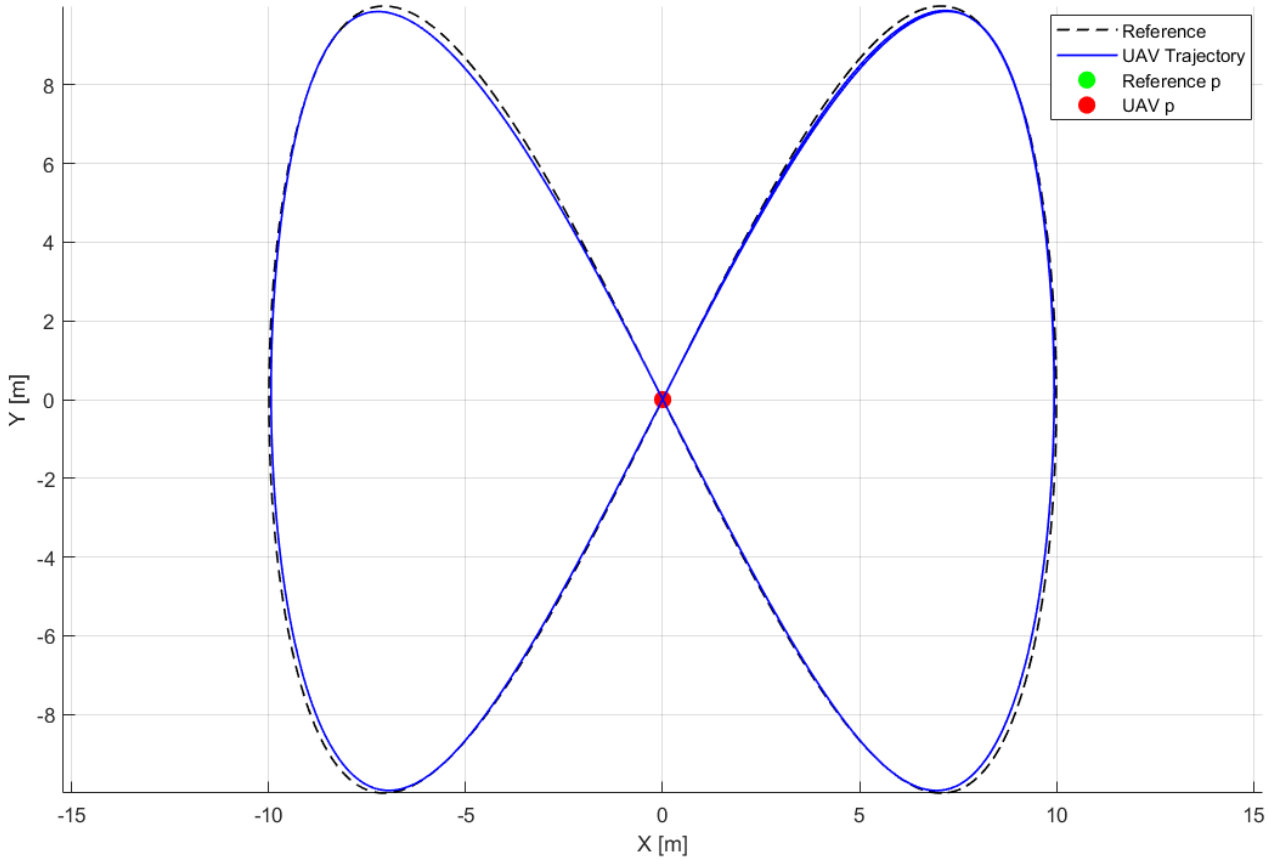
- Z velocity step behavior $K_{pos} = 1$
- Settling time (2%) ≈ 2.1 s (Nominal 2 s)
- Overshoot ≈ 1 % (Nominal 0 %)

2.3. Control architecture for position-heading tracking control
Full model test – 8 Shape

Reference trajectory

- X width = 10 meters
- Y width = 10 meters
- ω = 0.20 rad/s

$$V_{mean} \approx 3\ m/s$$
$$\Delta V \approx 1.5\ m/s$$
$$\text{RSME} = 4\ \text{m}$$

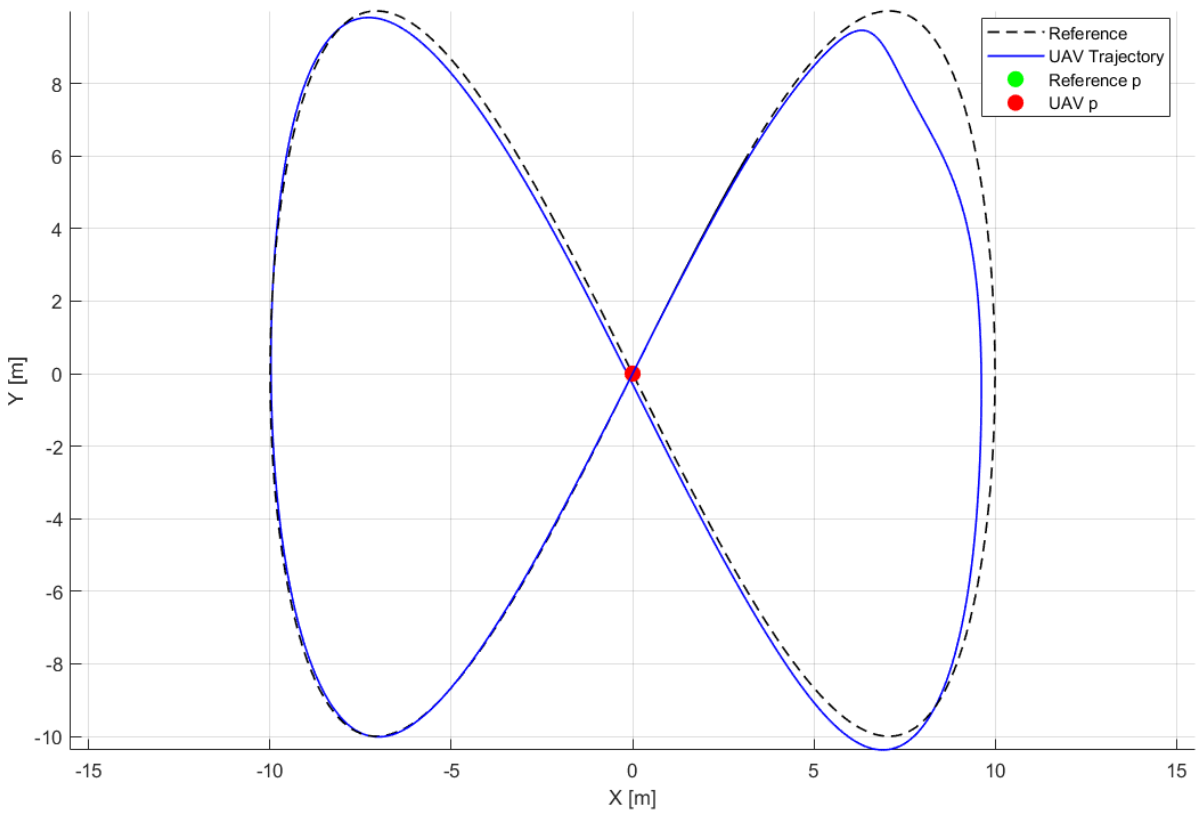


2.3. Control architecture for position-heading tracking control
Full model test – 8 Shape

Propeller's effectiveness reduction

A **reduction of a 20%** of effectiveness of one propeller at the first right turn of the 8 shape.

$$\mathrm{RSME} = 3.9\ \mathrm{m}$$



2.3. Control architecture for position-heading tracking control
Full model test – trajectory

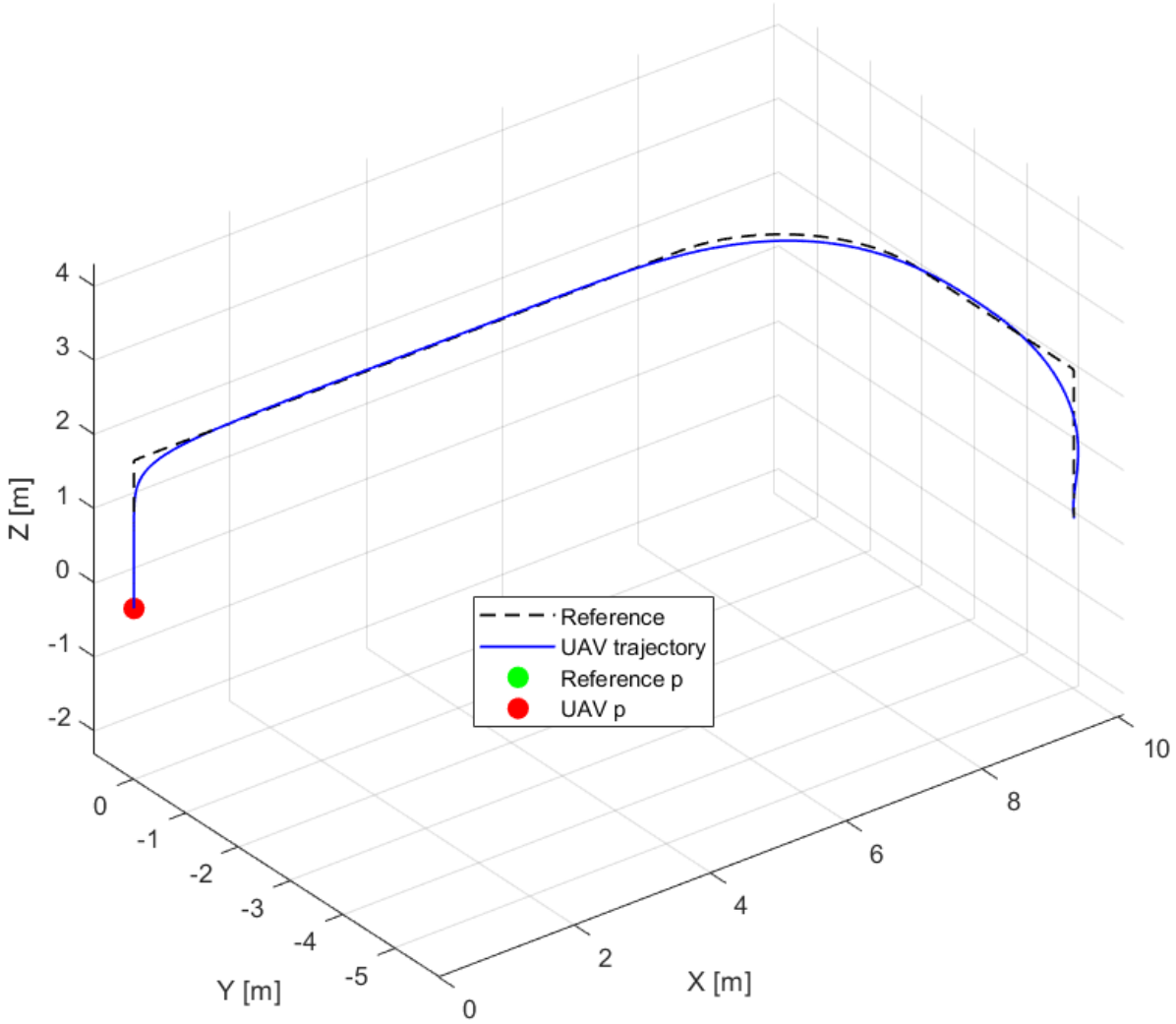
Reference trajectory

1. 2 meters ascend
2. X displacement of 8 meters
3. Turn on Y (R=2 meters)
4. 3 meters straight
5. 2 meters descend

$$V_{mean} \approx 1.5\ m/s$$

$$RSME\ =\ 1\ m$$

$$T_t = 14\ s$$



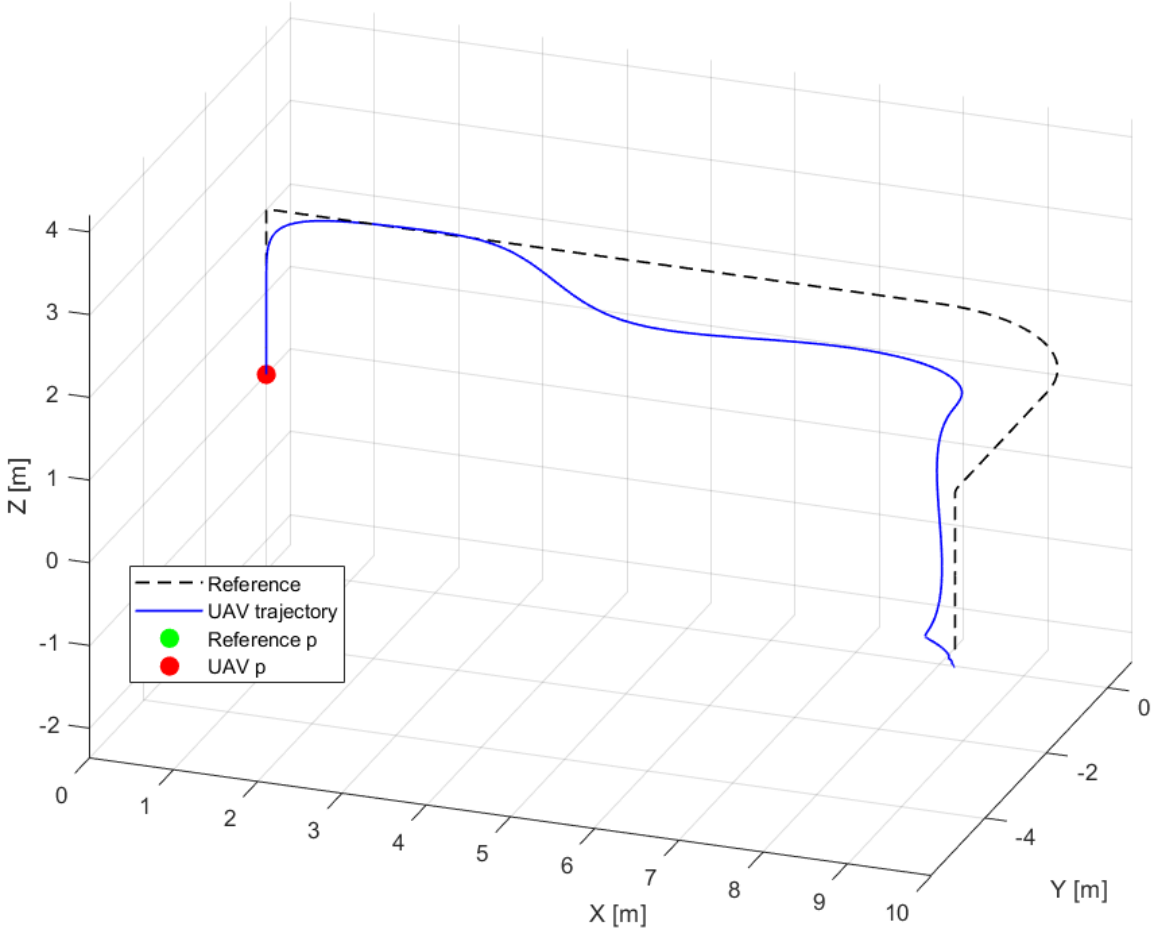
2.3. Control architecture for position-heading tracking control
Full model test – trajectory

Propeller's effectiveness reduction

A reduction of a 20% of effectiveness of one propeller after the initial vertical ascent.

Failure leads to error on Y tracking of 1 meter and a velocity lost that leads to continuous error that deteriorates the path.

$$\mathrm{RSME} = 1.3\ \mathrm{m}$$
$$T_t = 14\ \mathrm{s}$$



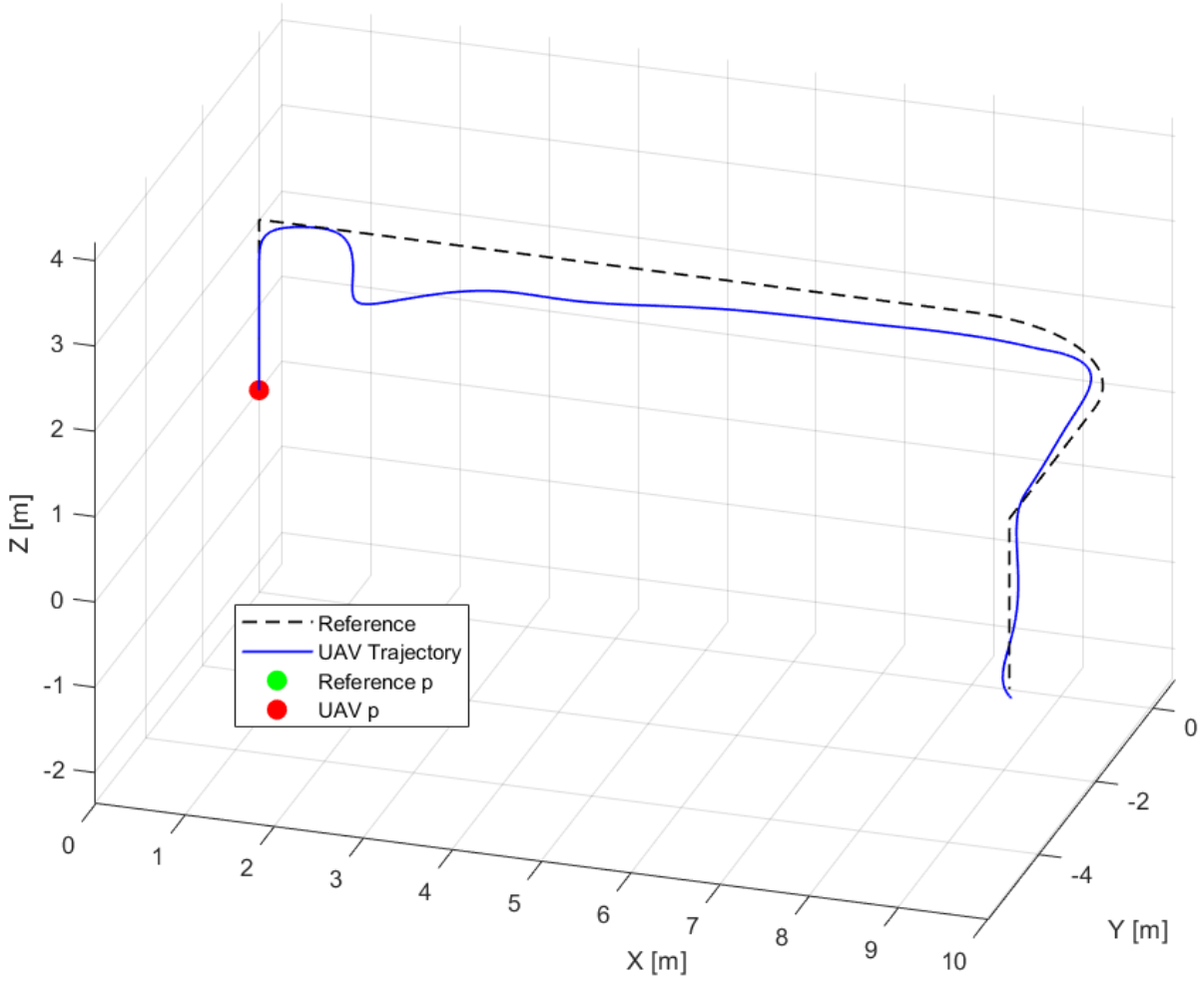
2.3. Control architecture for position-heading tracking control
Full model test – trajectory

Propeller's effectiveness reduction

A reduction of a 20% of effectiveness of one propeller after the initial vertical ascent.

Reference configured to not get ahead some distance of the UAV to improve its behavior while doubling time taken to finish the path.

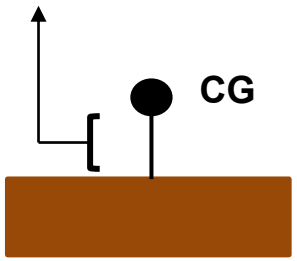
$$\text{RSME} = 1.1\ \text{m}$$
$$T_t = 30\ \text{s}$$



3. Adaptive Augmentation

3.1. Adaptive Augmentation
Package definition

$UAV.r_c \in [0.04, 0.06]\ m$



$UAV.m_p \in [0, 1]$ kg

EQUATIONS MODIFICATIONS

1. $UAV.m = UAV.m_0 + UAV.m_p$
2. $J_{p,CG} = UAV.m_p * UAV.r_c^2$
3. $J = J + [100; 010; 000]J_{p,CG}$

UAV New Behavior

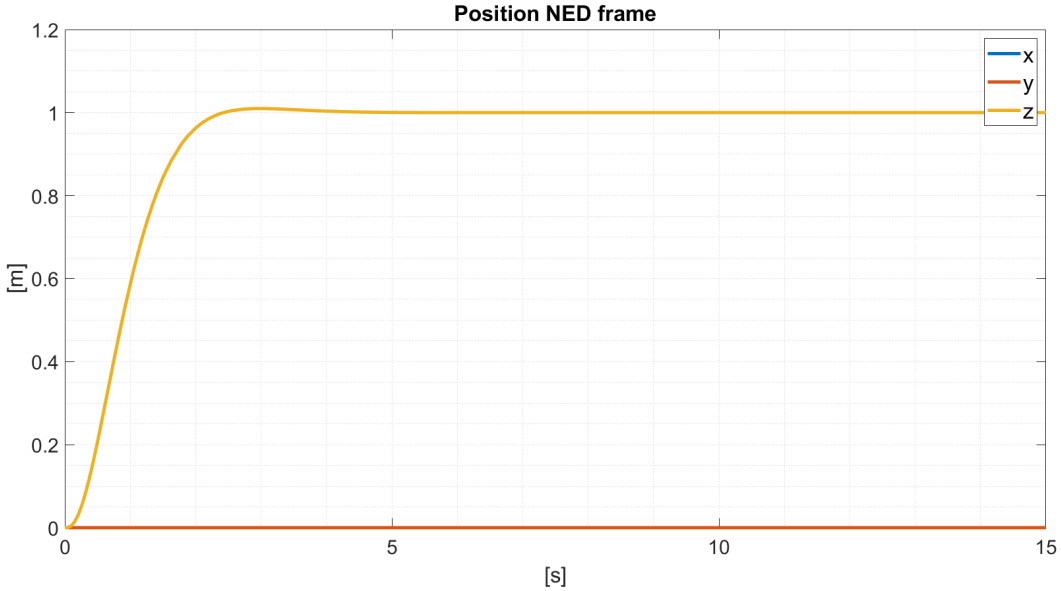
$K_{i,z}$ in the Z position controller was set null in the nominal design due to **complete knowledge of the UAV mass**. In order to correct the Z bias due to the package, **activation of this term is needed**. $K_{i,xy}$ is taken as reference.

J change is around 10% of total in XY rotations, not such a big change for not aggressive maneuvers.

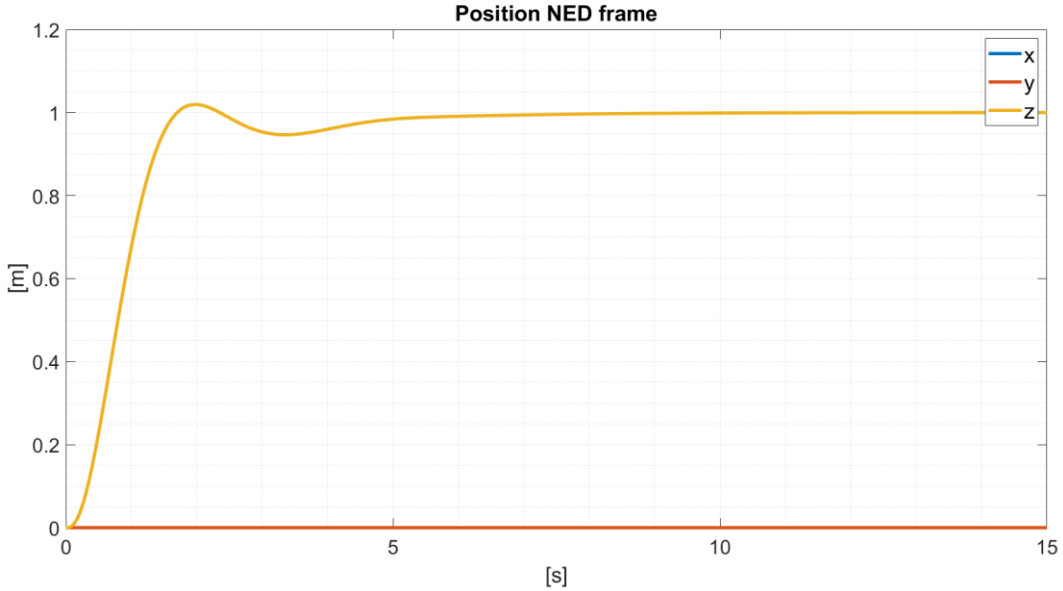
$\Longrightarrow$ $K_{i,z} = 2$, converges to 5% In 10 seconds. High values **worsen transitory**.

3.1. Adaptive Augmentation

Package definition – Z position control update



- Z velocity step behavior $K_{\mathrm{I}} = 0$
- Settling time (2%) $\approx 2.1\ \mathrm{s}$
- Overshoot $\approx\ 1\ \%$



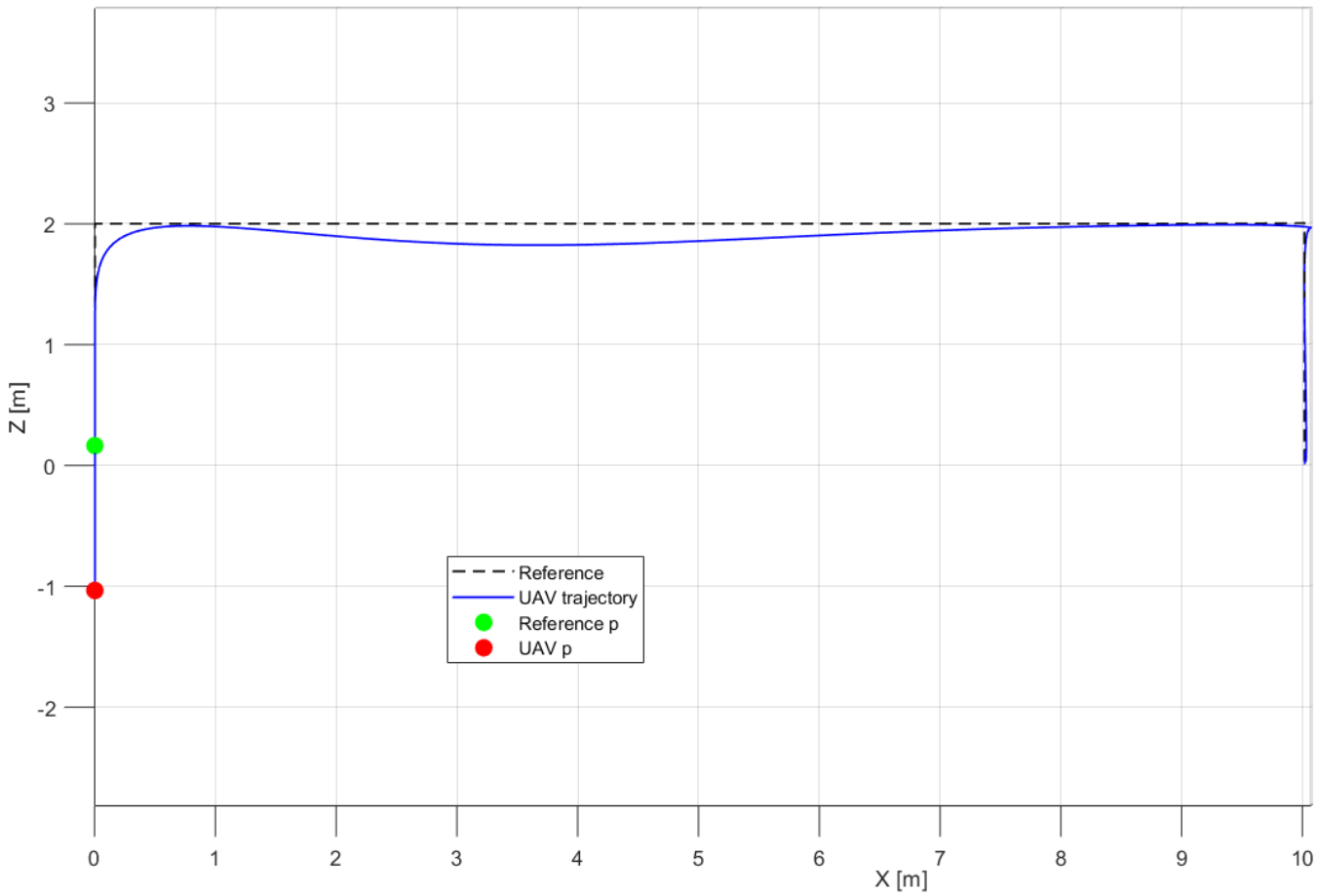
- Z velocity step behavior $K_{\mathrm{I}} = 2$
- Settling time (2%) $\approx 4.7\ \ s$
- Overshoot $\approx\ 2\ \%$
- Undershoot $\approx\ 6\ \%$

3.1. Adaptive Augmentation
Package definition - Tracking

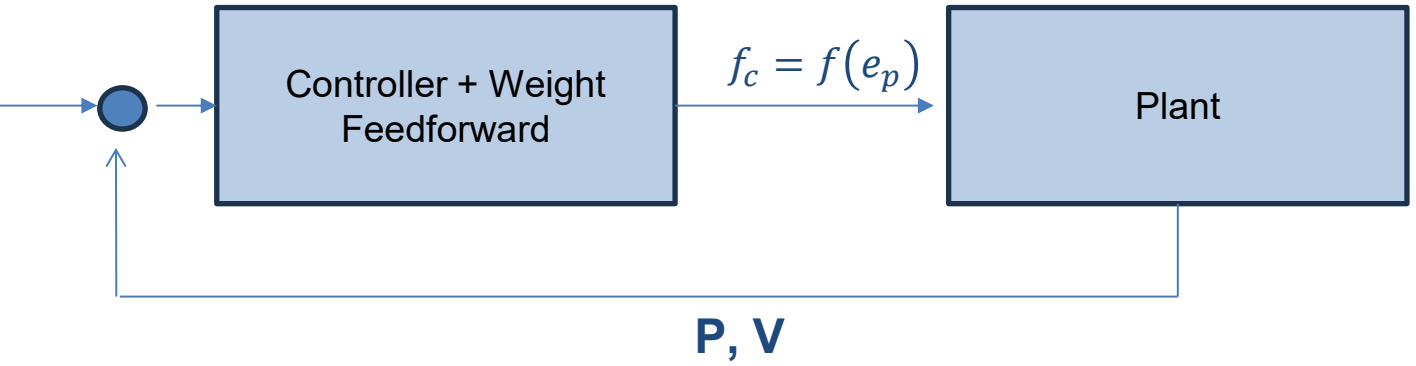
Trajectory tracking with package

Due to increase in $K_{i,z}$, trajectory tracking is reasonably good, but Z position error peaks initially around -1 meters and any vertical commands has light error due to change in Z controller.

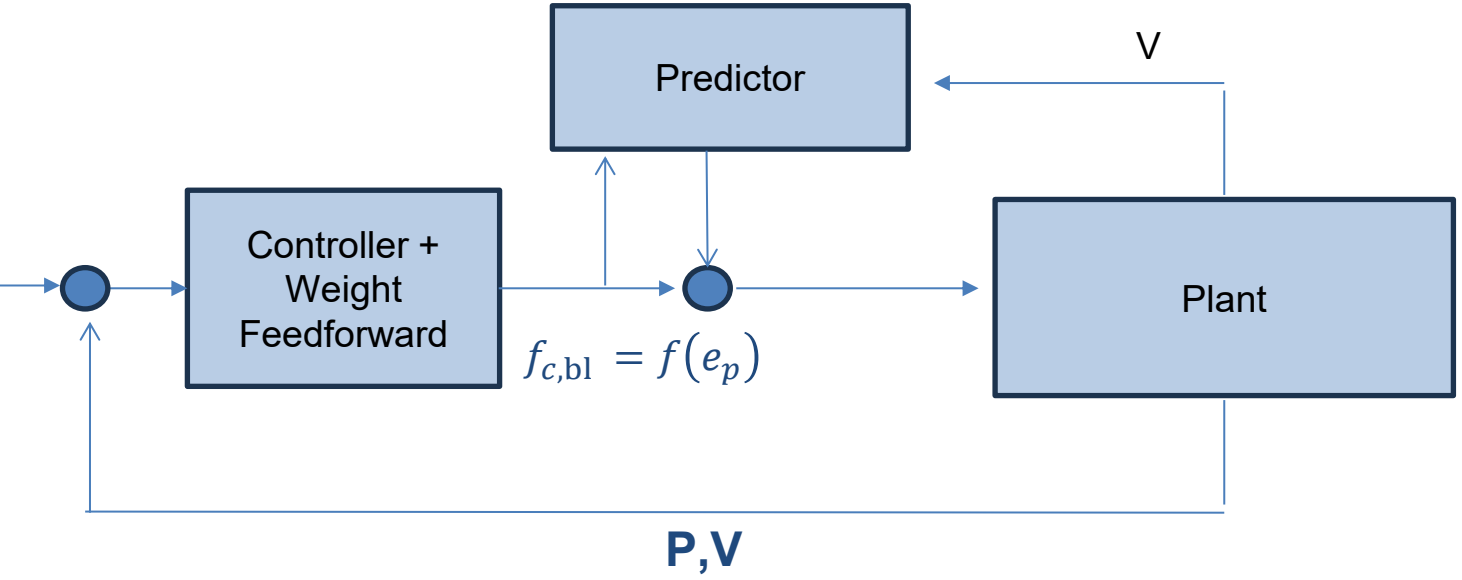
$$\text{RSME} = 1.1 \text{ m}$$
$$T_t = 14 \text{ s}$$



3.2. PBMRAC
Reason and Objectives



Slow convergence, need for **integral action to achieve equilibrium**, with limitations in its value due to transient requirements.



With the initial estimation of the parameters, the predictor **compares the internal simulation** results (with the baseline controller as input) **with the real state** of the UAV, **updates** the values of the **estimations** and ads a controller force that **complements the baseline** one to vanish the predictor error. Does not need to know the structure of the baseline controller and can be unplugged if necessary.

3.2. PBMRAC
Equations

Parameter uncertainties

1. $UAV.m$
2. $\boldsymbol{D}$
3. Bias (Wind)
4. $\boldsymbol{\Lambda_f}$

Plant equations

$$m\dot{\boldsymbol{v}}_i = -mg\boldsymbol{e_3} + \boldsymbol{f_e} + \boldsymbol{\Lambda_f f_c}$$

$$\boldsymbol{f_e} = -\boldsymbol{D}(\boldsymbol{v} - \boldsymbol{v_{wind}})$$

Predictor

$$\dot{\boldsymbol{v}}_p = \boldsymbol{A_p v_p} + \boldsymbol{B_p}\left(\boldsymbol{\Lambda u} + \boldsymbol{\varphi}(\boldsymbol{x_p})^\top \boldsymbol{\theta_p}\right) - L\hat{\boldsymbol{e}}$$

$$\dot{\boldsymbol{v}}_p = \boldsymbol{A_p v_p} + \boldsymbol{B_p}\left(\boldsymbol{\varphi_a}\left(\boldsymbol{x_p}, \boldsymbol{u}\right)^\top \boldsymbol{\theta_a}\right) - L\hat{\boldsymbol{e}}$$

$\boldsymbol{A_p}$ & $\boldsymbol{B_p}$ known, all uncertainties (constant) in $\boldsymbol{\Theta_a}$ and their regressor in $\boldsymbol{\varphi_a}$. Term $L\hat{\boldsymbol{e}}$ works on predictor and UAV dynamics to converge.

$$A_p = 0$$

$$B_p = 1/\bar{m}$$

$$\varphi_a^T = \begin{pmatrix} u_x & 0 & 0 & 1 & 0 & 0 & -v_1 & 0 & 0 \\ 0 & u_y & 0 & 0 & 1 & 0 & 0 & -v_2 & 0 \\ 0 & 0 & u_z & 0 & 0 & -1 & 0 & 0 & -v_3 \end{pmatrix}$$

$$\theta^T = [\Lambda_x/m, \Lambda_y/m, \Lambda_z/m, \theta_x, \theta_y, \bar{m}g, \bar{m}/mD_x, \bar{m}/mD_y, \bar{m}/mD_z];$$

3.2. PBMRAC

Equations

Predictor error dynamic

$$\dot{\hat{\boldsymbol{e}}}_{\boldsymbol{p}} = \dot{\boldsymbol{v}}_{\boldsymbol{p}} - \dot{\hat{\boldsymbol{v}}}_{\boldsymbol{p}} = \boldsymbol{B}_{\boldsymbol{p}}\left(\underbrace{\left(\boldsymbol{\Lambda} - \hat{\boldsymbol{\Lambda}}\right)}_{\boldsymbol{\Delta\Lambda}}\boldsymbol{u} + \boldsymbol{\varphi}_{\boldsymbol{p}}\left(\boldsymbol{x}_{\boldsymbol{p}}\right)^{\top}\underbrace{\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}\right)}_{\boldsymbol{\Delta\theta}}\right) - L\hat{\boldsymbol{e}} \longrightarrow \dot{\hat{\boldsymbol{e}}}_{p} = \boldsymbol{B}_{\boldsymbol{p}}\left(\boldsymbol{\varphi}_{\mathrm{a}}\left(\boldsymbol{x}_{\boldsymbol{p}}, \boldsymbol{u}\right)^{\top}\boldsymbol{\Delta\theta}_{\boldsymbol{a}}\right) - L\hat{\boldsymbol{e}}$$

- **Lyapunov function proposed** $(\boldsymbol{\Gamma}_{\boldsymbol{a}} > 0)$:

$$V(\hat{\boldsymbol{e}}, \boldsymbol{\Delta\theta}_{\boldsymbol{a}}) := \frac{1}{2}|\hat{\boldsymbol{e}}|^2 + \frac{1}{2}\Delta\boldsymbol{\theta}_{\boldsymbol{a}}^{\top}\boldsymbol{\Gamma}_{\boldsymbol{a}}\Delta\boldsymbol{\theta}_{a} \qquad \mathcal{L}_{\mathbf{f}}V = -\hat{\boldsymbol{e}}^{\top}L\hat{\boldsymbol{e}} + \Delta\boldsymbol{\theta}_{\boldsymbol{a}}^{\top}\left(\boldsymbol{\varphi}_{\boldsymbol{a}}\left(\boldsymbol{u}, \boldsymbol{x}_{\boldsymbol{p}}\right)\boldsymbol{B}_{p}^{\top}\hat{\boldsymbol{e}} - \boldsymbol{\Gamma}_{\boldsymbol{a}}^{-1}\dot{\hat{\boldsymbol{\theta}}}_{\boldsymbol{a}}\right)$$

For Lie Derivative of V to be **negative semidefinite**, the **parenthesis** needs to be **null:**

$$\dot{\hat{\boldsymbol{\theta}}}_{\boldsymbol{a}} = \boldsymbol{\Gamma}_{a}\boldsymbol{\varphi}_{a}\left(\boldsymbol{u}, \boldsymbol{x}_{p}\right)\boldsymbol{B}_{p}^{\top}\hat{\boldsymbol{e}}$$

Gain update law to assure $\hat{\boldsymbol{e}} \to 0$

3.2. PBMRAC
Equations

Adaptive input u_a

$$\boldsymbol{u} = \boldsymbol{u}_b + \boldsymbol{u}_a$$

$$\boldsymbol{u_a} = \left(\widehat{\boldsymbol{\Lambda}}^{-1} - \boldsymbol{I}\right)\boldsymbol{u_b} - \widehat{\boldsymbol{\Lambda}}^{-1}\boldsymbol{\varphi}_p\left(\boldsymbol{x}_p\right)^\top\widehat{\boldsymbol{\theta}}_p \quad \rightarrow \quad \boldsymbol{u} = \widehat{\boldsymbol{\Lambda}}^{-1}\left(\boldsymbol{u}_b - \boldsymbol{\varphi}_p\left(\boldsymbol{x_p}\right)^\top\widehat{\boldsymbol{\theta}}_p\right)$$

$$\dot{\hat{\boldsymbol{v}}}_p = \boldsymbol{A}_p\boldsymbol{v}_p + \boldsymbol{B}_p\left(\widehat{\boldsymbol{\Lambda}}\left(\widehat{\boldsymbol{\Lambda}}^{-1}\boldsymbol{u}_b - \boldsymbol{\varphi}_p\left(\boldsymbol{x}_p\right)^\top\widehat{\boldsymbol{\theta}}_p\right) + \boldsymbol{\varphi}_p\left(\boldsymbol{x}_p\right)^\top\widehat{\boldsymbol{\theta}}_p\right) + L\hat{\boldsymbol{e}} \;=\; \boldsymbol{A}_p\boldsymbol{v}_p + \boldsymbol{B}_p\boldsymbol{u}_b + L\hat{\boldsymbol{e}}$$

For $\hat{\boldsymbol{e}} \rightarrow 0$, predictor behavior is identical to the plant

Why no attitude augmentation?

Expected modification of attitude parameters is not as high as the position ones. However, the **presence of expected bias** in the behavior , i.e. a displacement in XY plane of the CG position, makes **impossible for the baseline attitude controller to converge** to the desired orientation due to **lack of integral action**, so it would be needed to model the attitude uncertainties in the predictor.

3.2. PBMRAC

Projector + e modification

Projector

Operator in the adaptive laws **that avoids drift of the gains**, **bounding them** in a predefined space and while keeping the Lie Derivative of Lyapunov function negative semidefinite.

e-modification

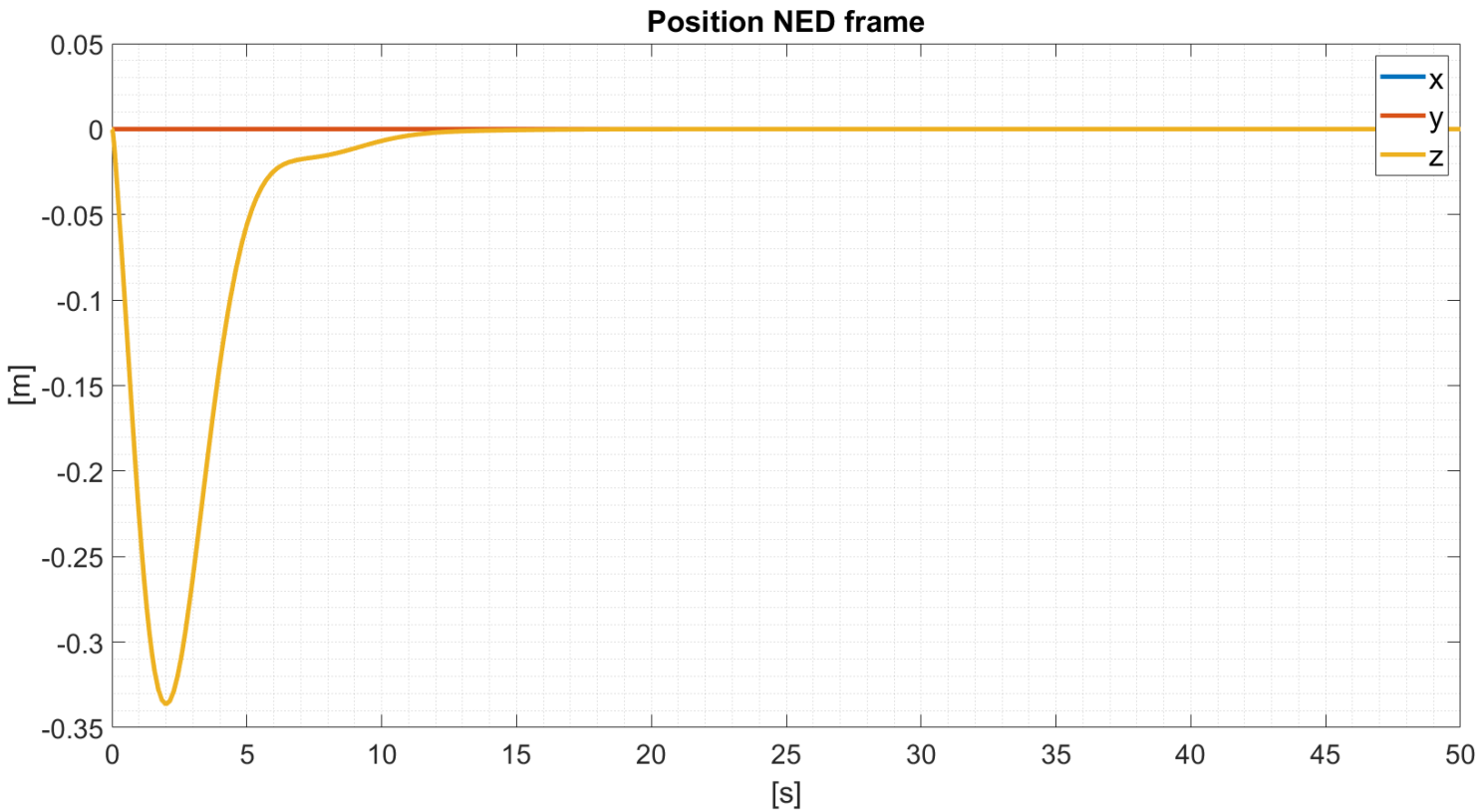
Modification to the adaptive laws that provides damping**, slowing down** the **evolution** of the gains.

3.2. PBMRAC

Static desired position test

Mass uncertainty

Assuming maximum package weight (1 kg).

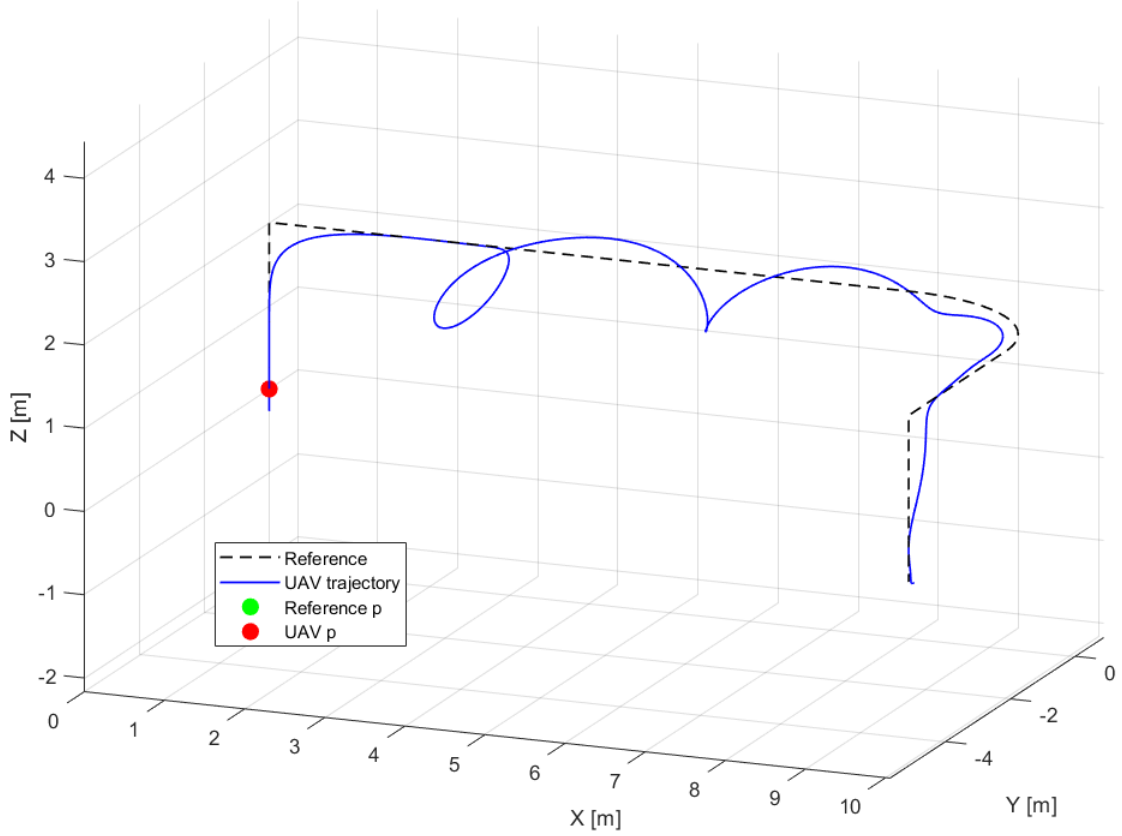


3.2. PBMRAC

Trajectory Test

Mass uncertainty + Propeller efficiency reduction

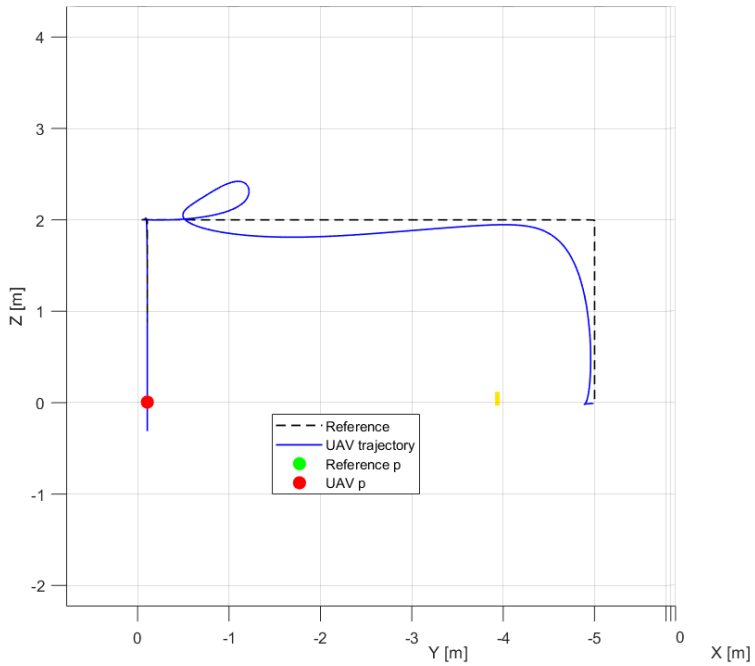
Mass uncertainty is matched in less than 3 seconds, with an initial descent of less than 20 centimeters (0.16 kg of Δm). Small package enough to avoid saturation.
After a transient with an overshoot of 2 meters, the UAV converges in around 5 seconds and recovers trajectory tracking performance.

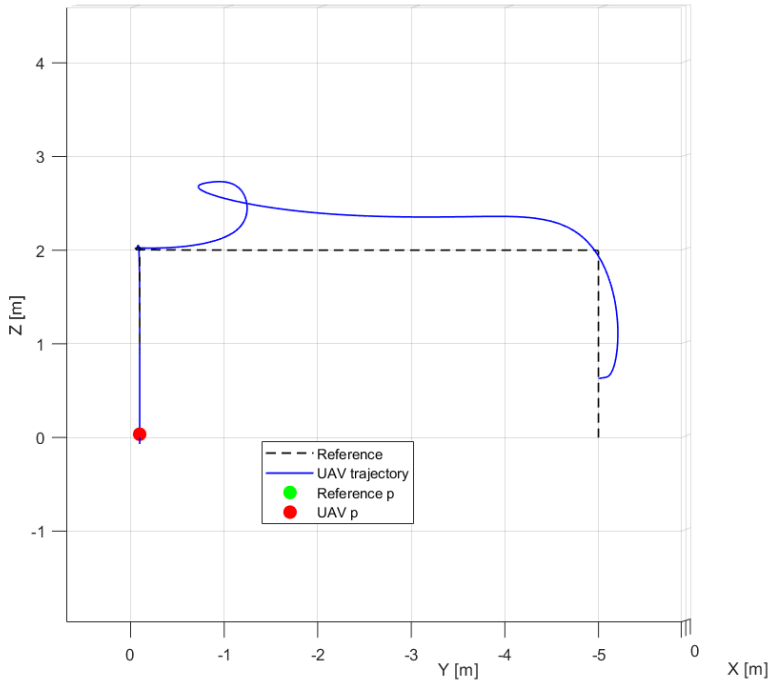


3.2. PBMRAC

Trajectory Test

Adaptive **ON**





Adaptive **OFF**

Wind of -5 m/s Y at end of ascend

Mass uncertainty not considered.

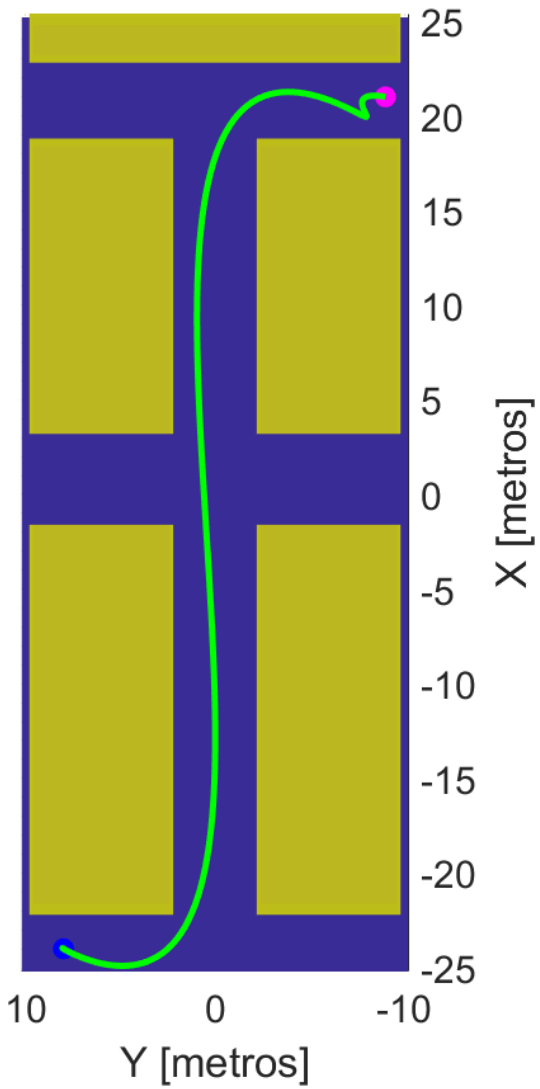
A **smaller overshoot and good convergence** of the behavior just before the right turn

4. Test delivery scenario

4. Test delivery scenario
Map & Trajectory

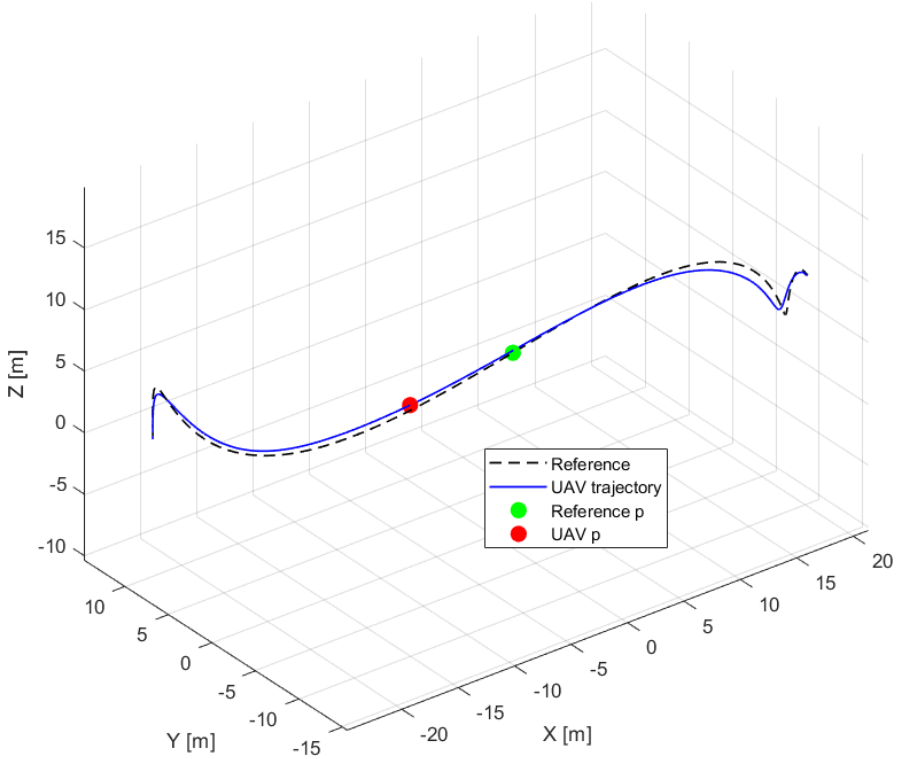
Trajectory around buildings of **characteristic length 10 meters**. Start in 2.5 meters height, with ascension to 5 meters and package delivery in 7.5 meters.
Total length $\approx 60\ \mathrm{m}$, reference achieves it in 22 seconds (around **2.7 m/s mean velocity**).
For the test, two considerations of **unforeseen events**:

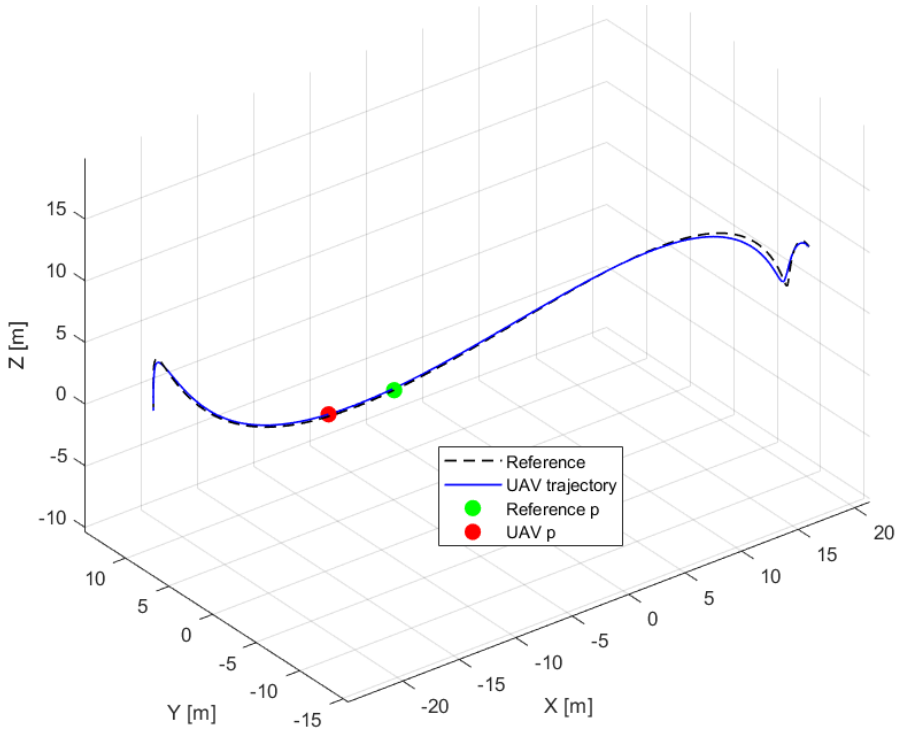
1. Crosswind in –Y direction, intensified in the crossroads.
2. Degradation of 10 % of all propellers' forces.



4. Test delivery scenario
Base Trajectory tracking

30 s



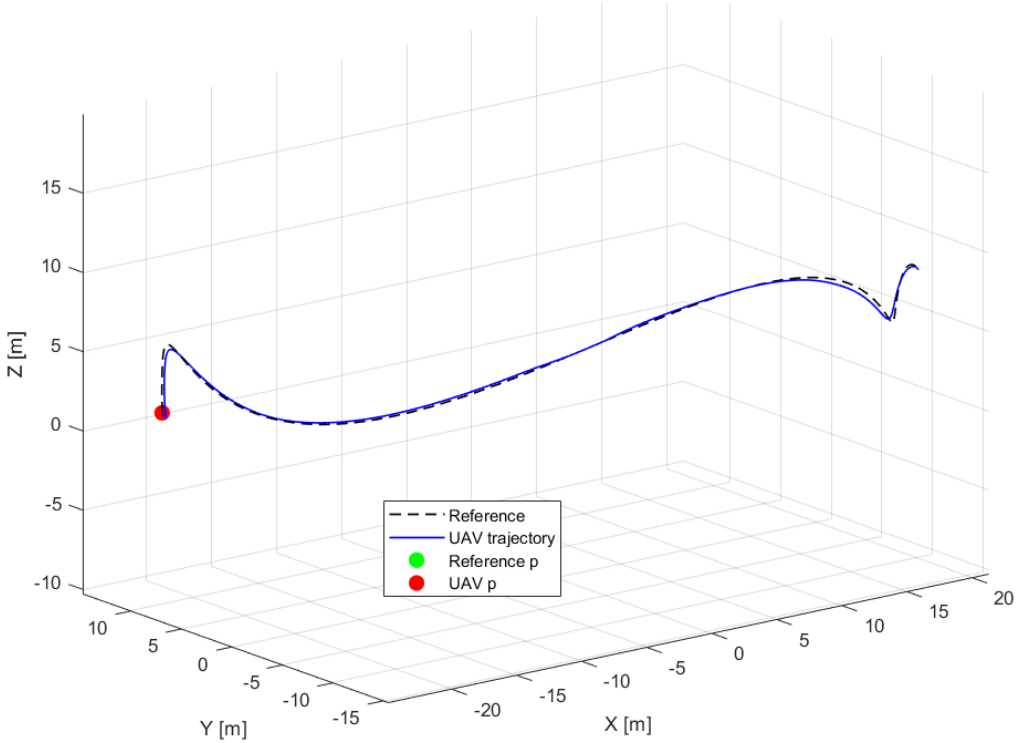


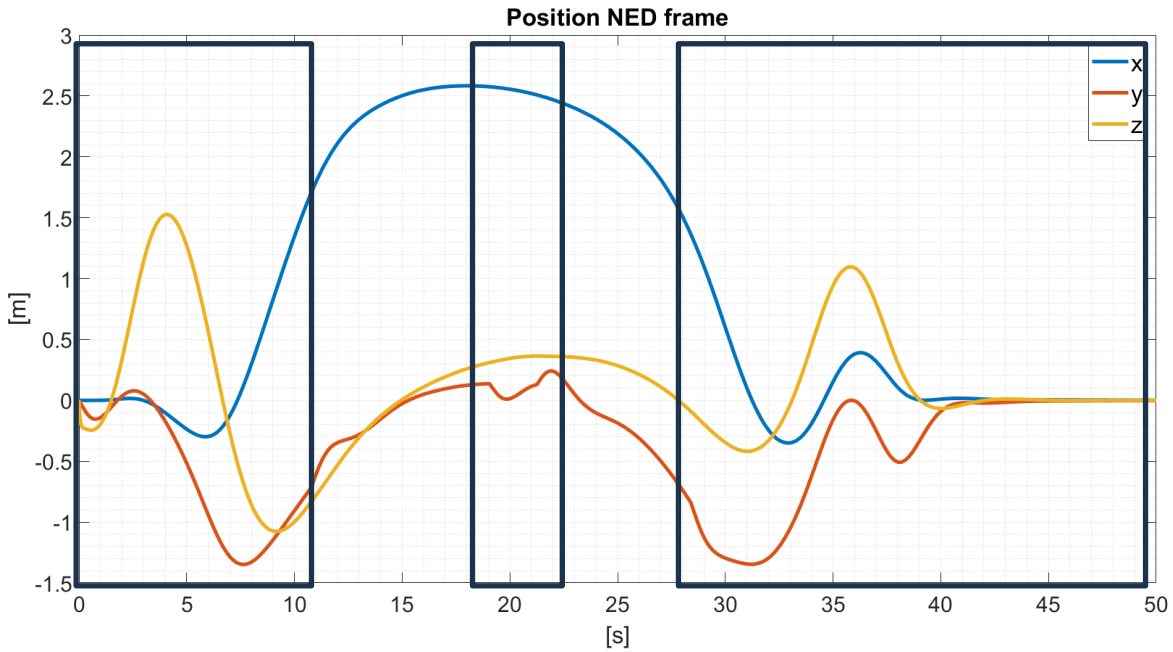
40 s

Mean distance of 1.4 meters between reference and UAV, with peaks of 9 meters in the straight (RMSE of 4.5 m).
Fulfilment of the trajectory in 30 seconds (2 m/s mean velocity).
For 40 seconds (slower reference), RMSE is 3.6 meters, with peaks of 6 meters in the straight.

4. Test delivery scenario
Trajectory tracking with wind

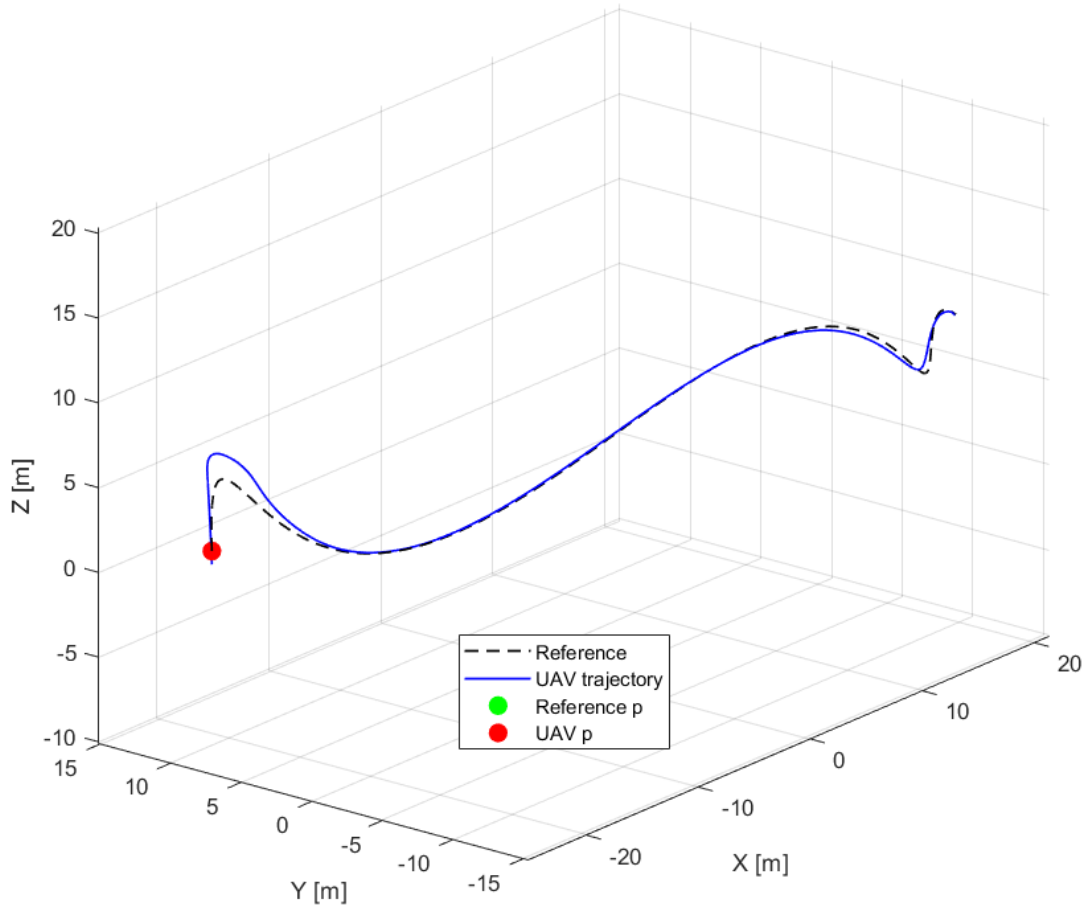
*To avoid saturation, a windspeed of 5 m/s (1 Newton force) has been considered in crossroads and 1 m/s in the rest.





4. Test delivery scenario

Trajectory tracking with wind & propeller effectiveness reduction



Reduction of all propellers' effectiveness in t = 0.

Overshoot in lambda estimation leads to overshoot in height, but fast convergence leads to no lateral displacement that could cause a collision of the UAV with surroundings.

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